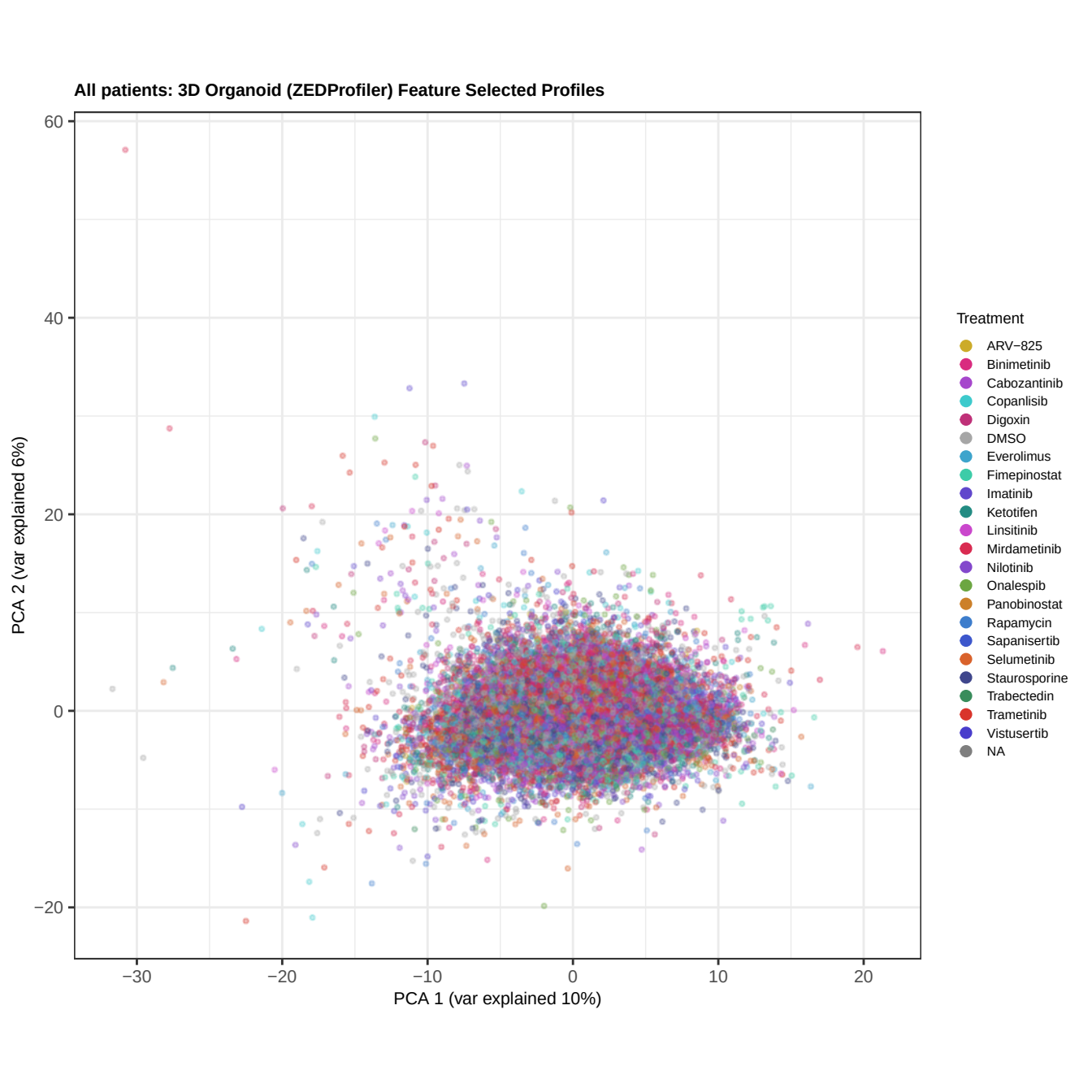
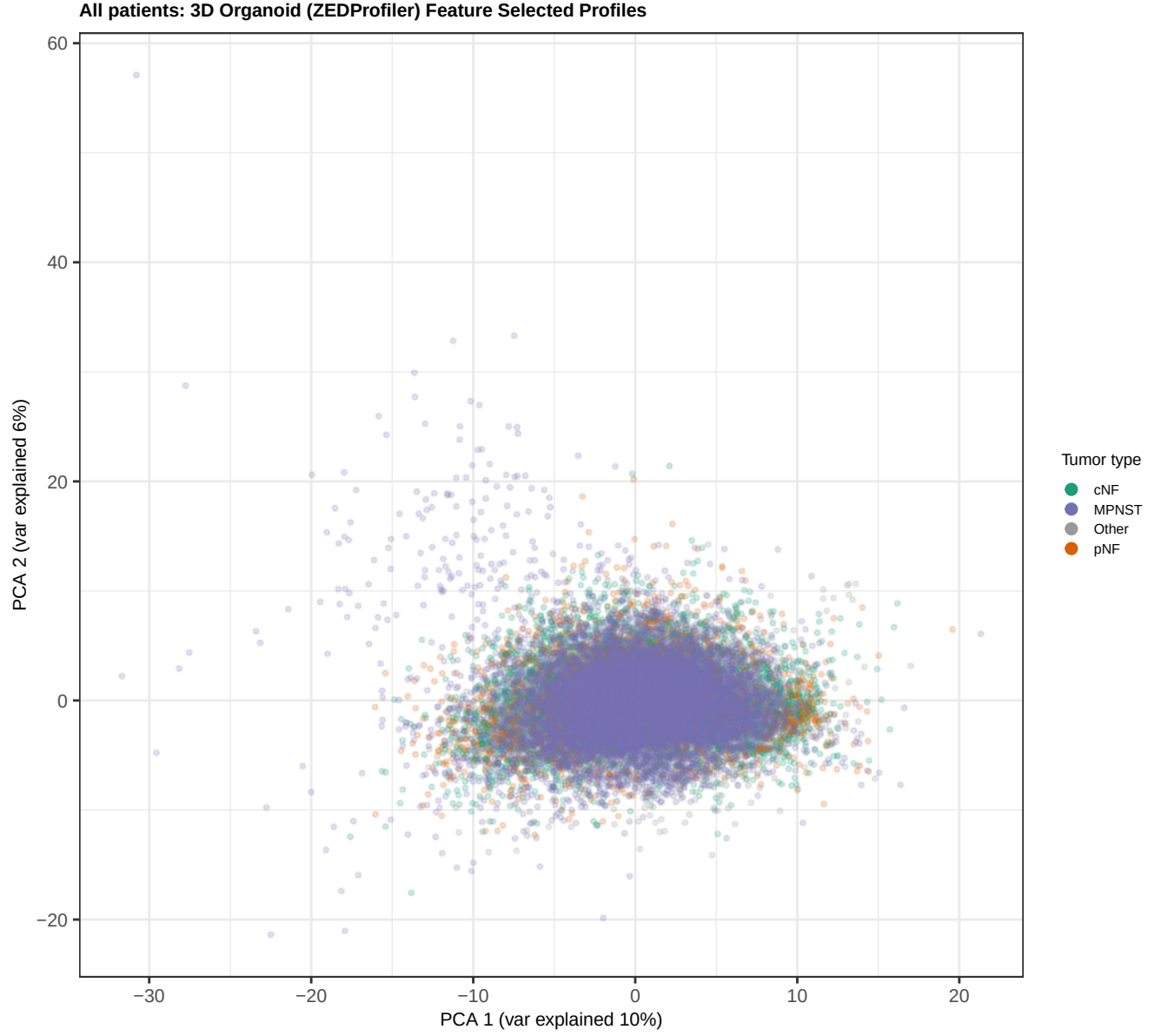
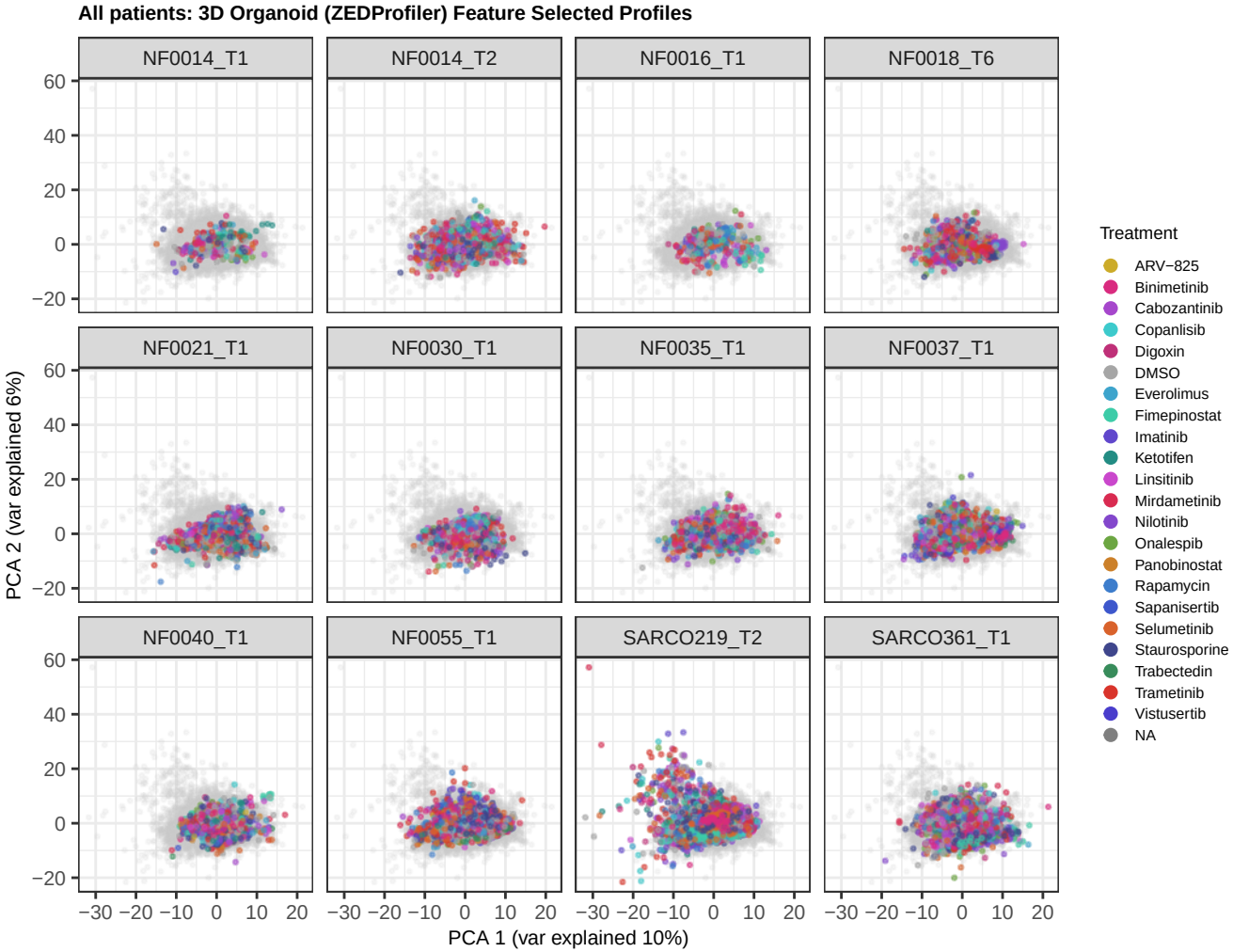
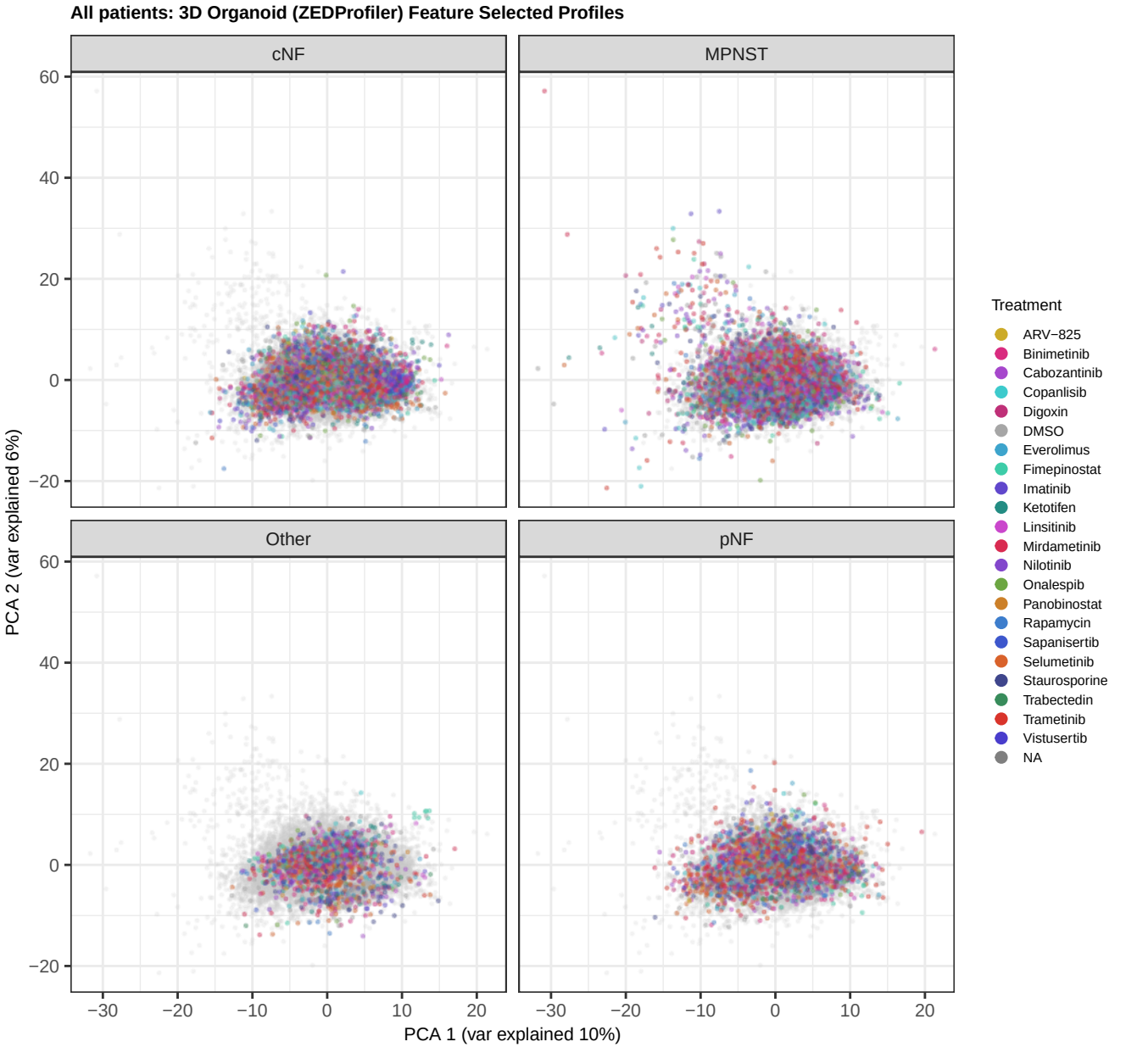
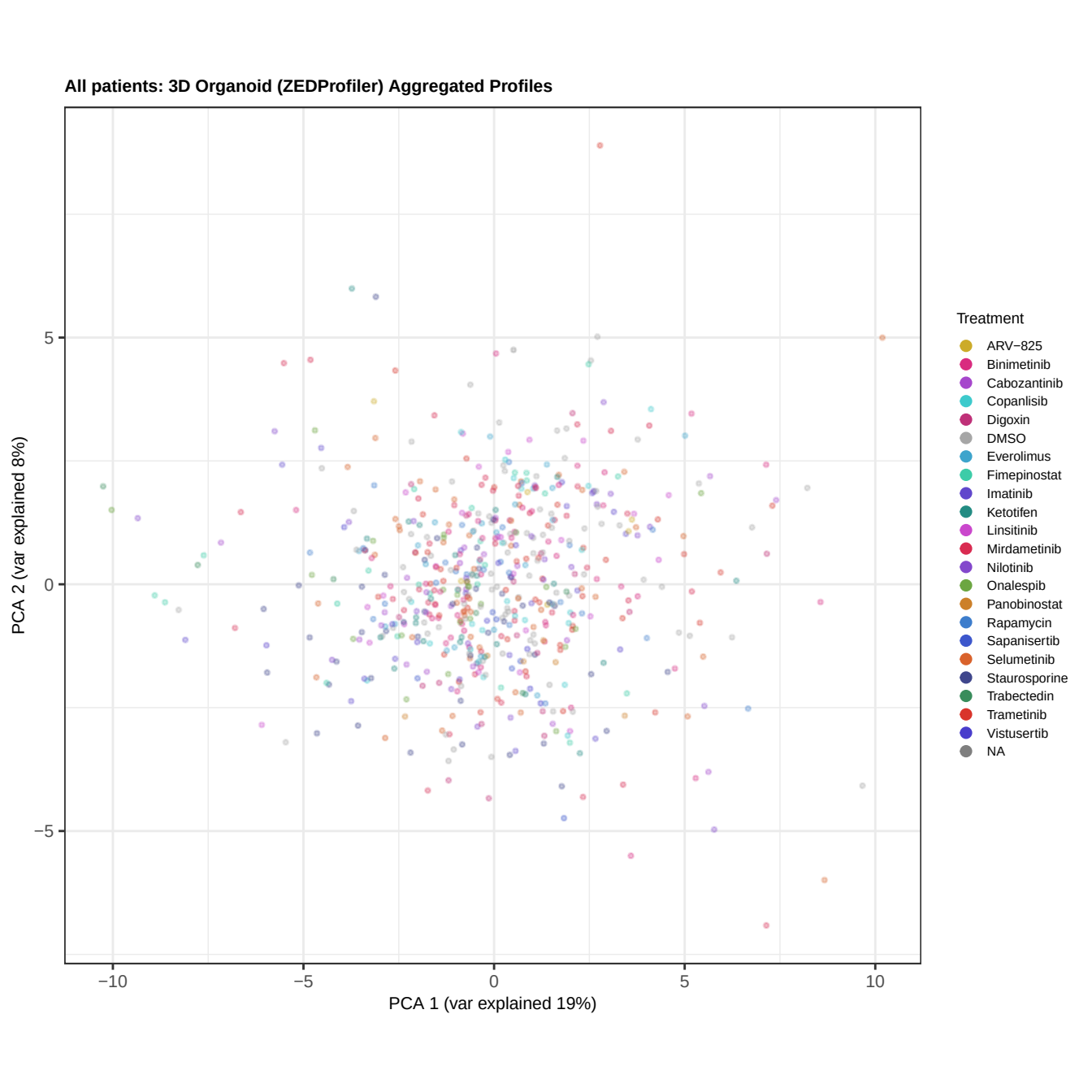
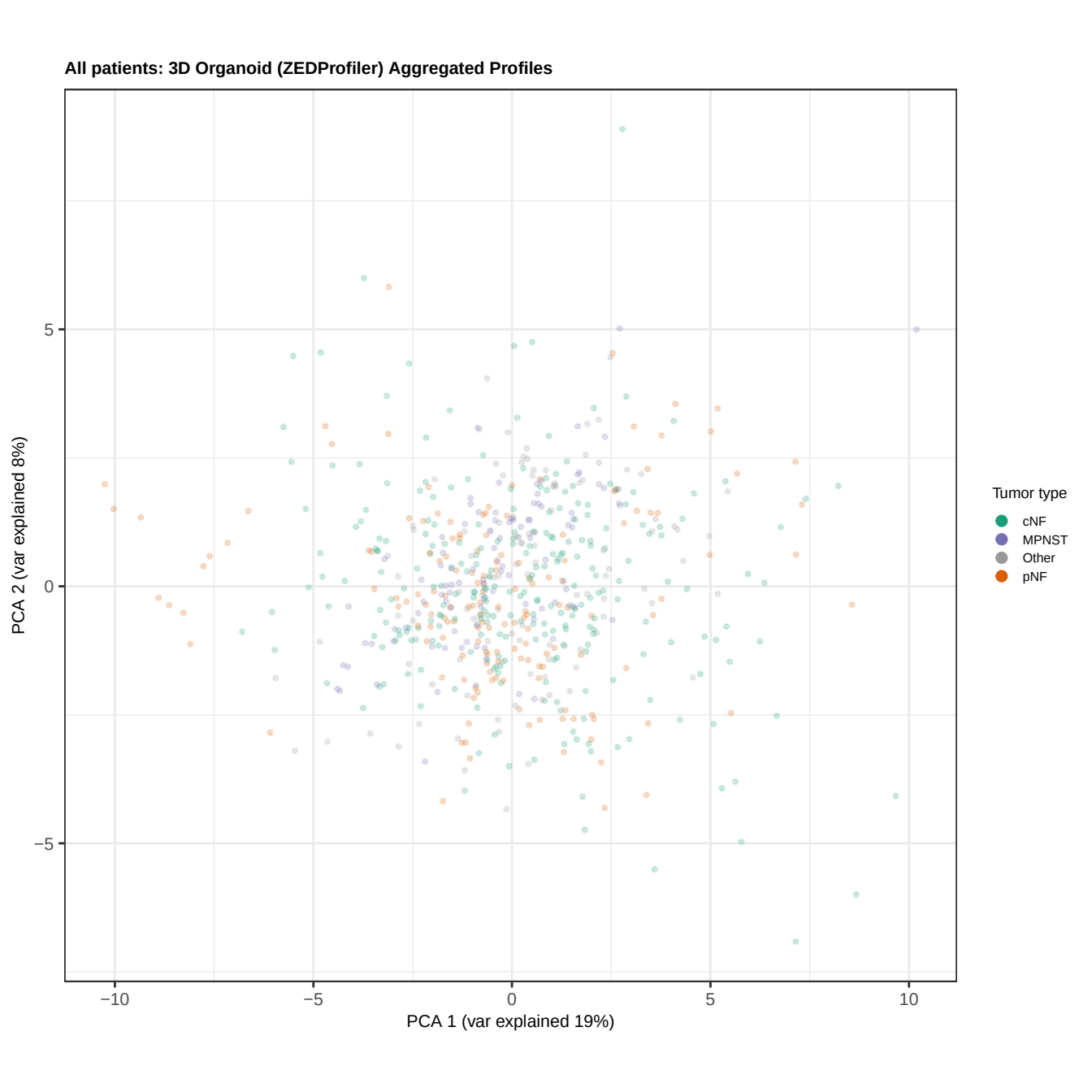

All patients: 3D Organoid (ZEDProfiler) Feature Selected Profiles
PCA 2 (var explained 6%)
PCA 1 (var explained 10%)
60
40
20
0
−20
−30
−20
−10
0
10
20
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA

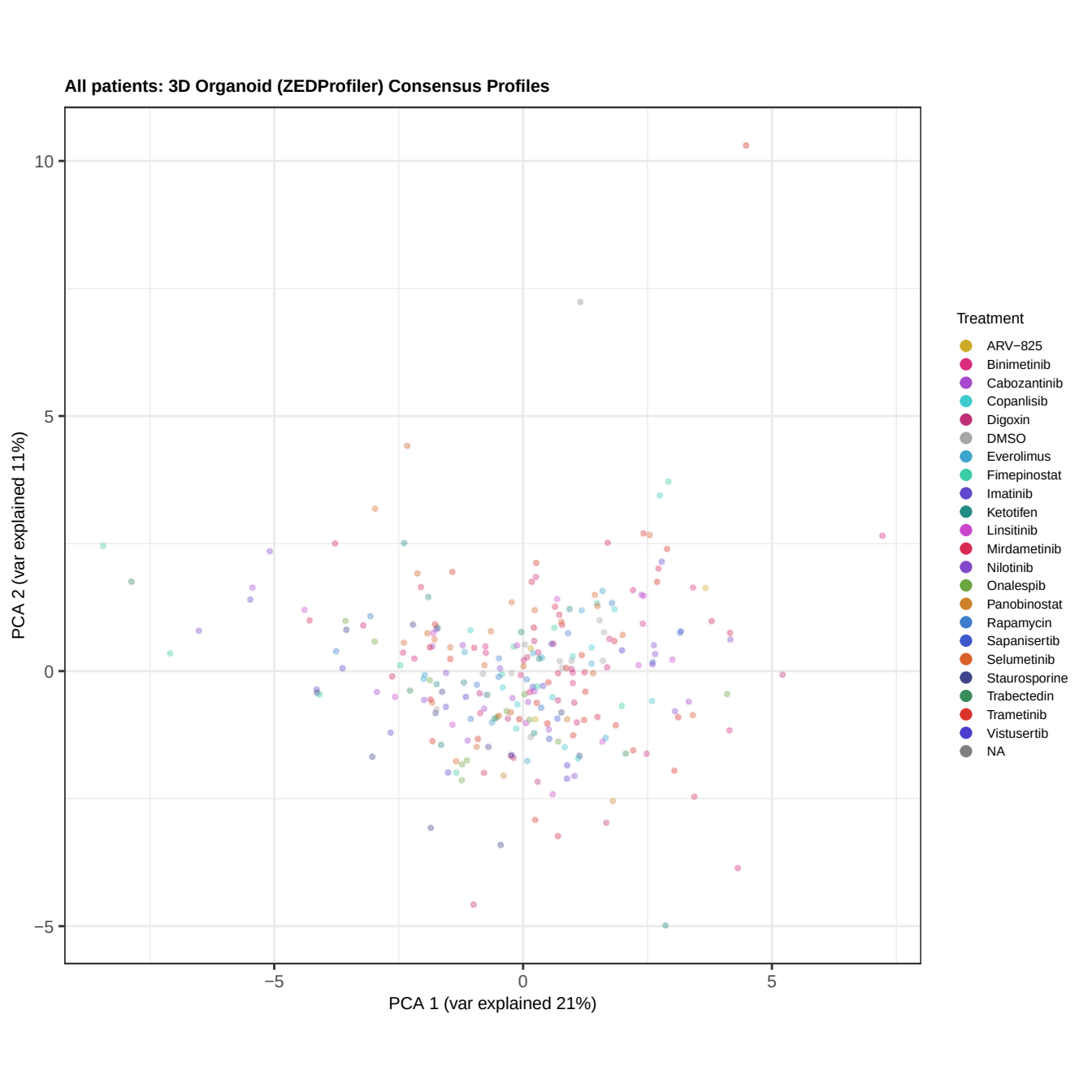

All patients: 3D Organoid (ZEDProfiler) Feature Selected Profiles
60
40
20
0
−20
PCA 2 (var explained 6%)
−30
−20
−10
0
10
20
PCA 1 (var explained 10%)
Tumor type
cNF
MPNST
Other
pNF

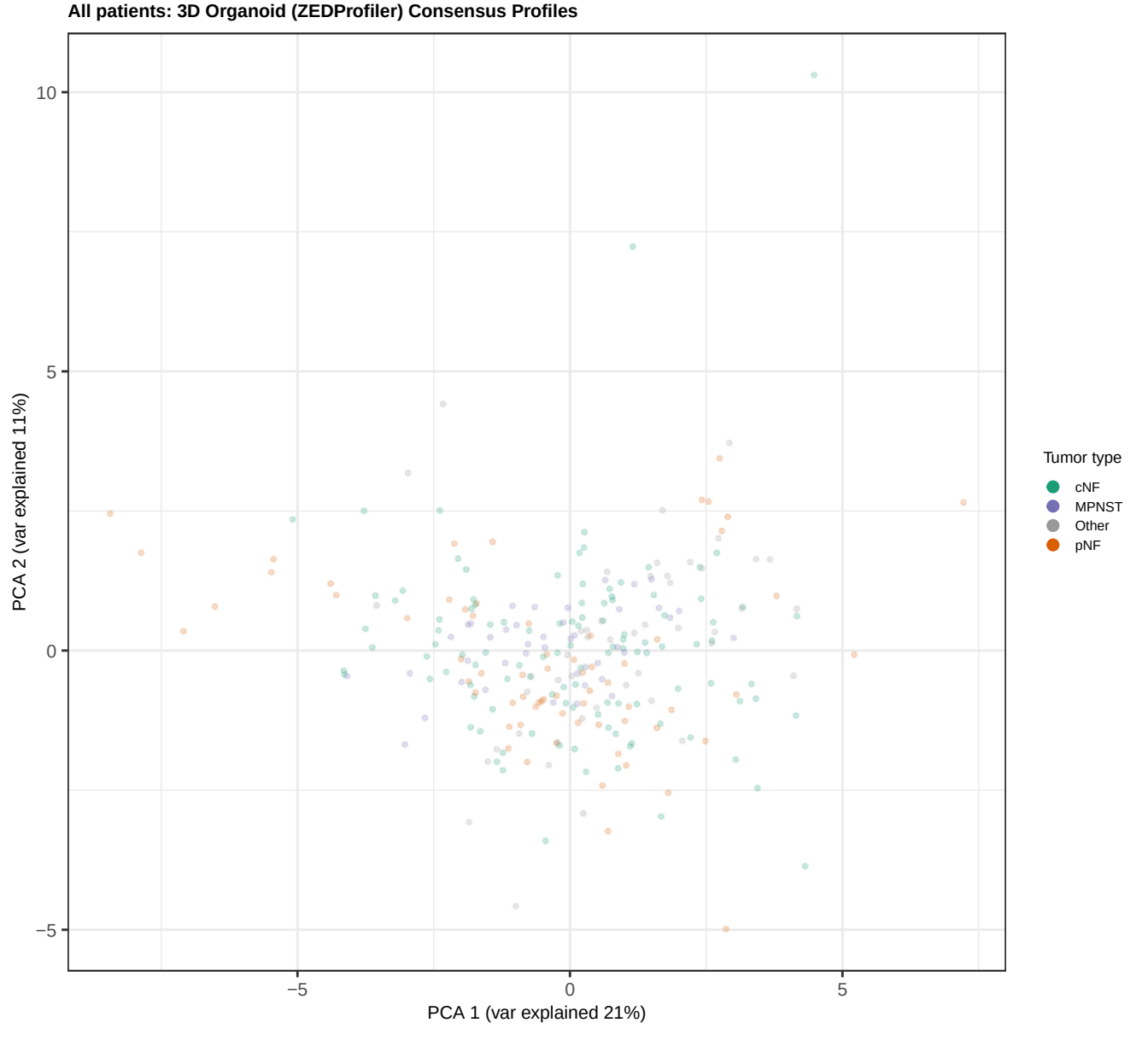

All patients: 3D Organoid (ZEDProfiler) Feature Selected Profiles
NF0014_T1
NF0014_T2
NF0016_T1
NF0018_T6
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1
PCA 2 (var explained 6%)
PCA 1 (var explained 10%)
60
40
20
0
−20
−30
−10
10
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA


All patients: 3D Organoid (ZEDProfiler) Feature Selected Profiles
cNF
MPNST
Other
pNF
PCA 2 (var explained 6%)
PCA 1 (var explained 10%)
60
40
20
0
−20
−30
−20
−10
0
10
20
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA


All patients: 3D Organoid (ZEDProfiler) Aggregated Profiles
PCA 2 (var explained 8%)
PCA 1 (var explained 19%)
5
0
−5
−10
−5
0
5
10
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA

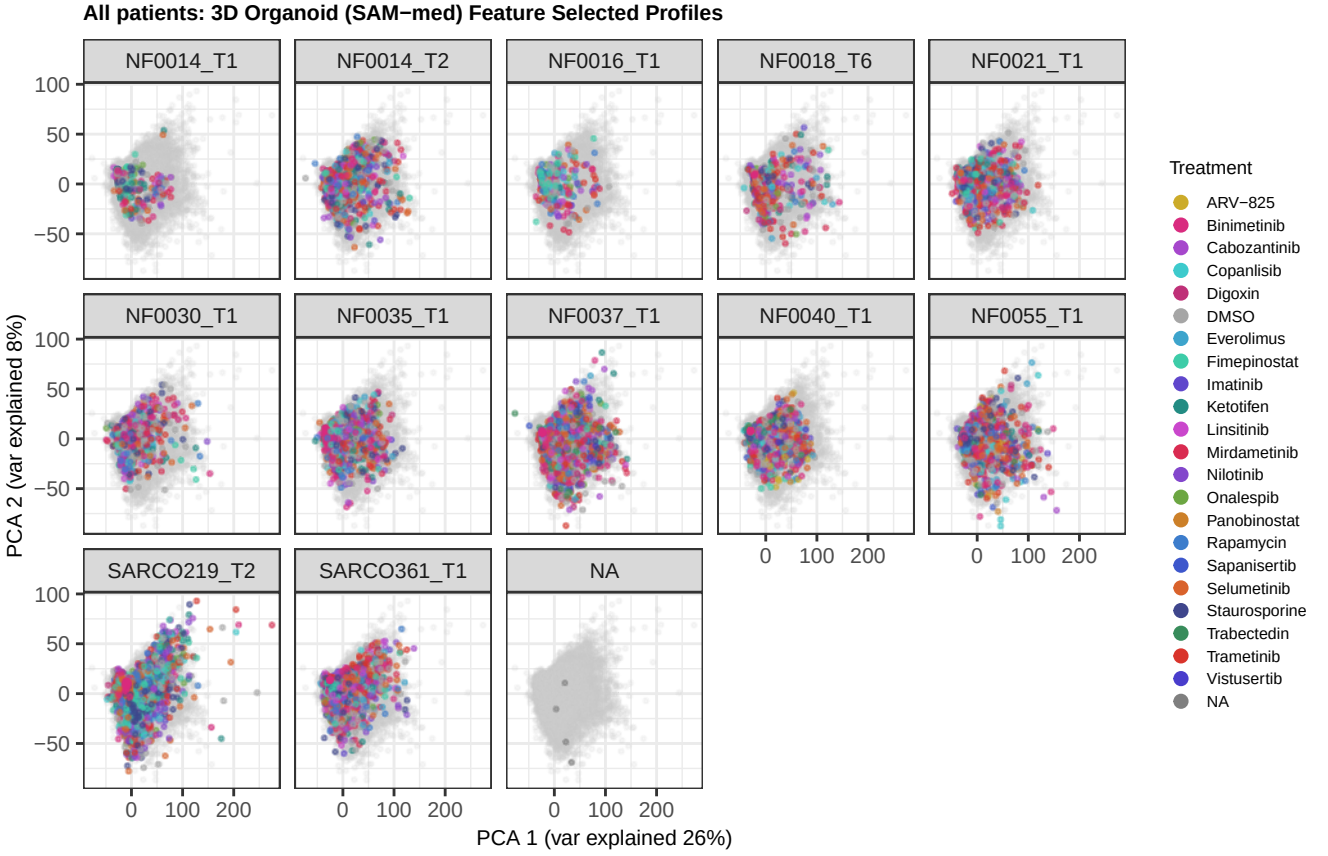

All patients: 3D Organoid (ZEDProfiler) Aggregated Profiles
PCA 2 (var explained 8%)
5
0
−5
−10
−5
0
5
10
PCA 1 (var explained 19%)
Tumor type
cNF
MPNST
Other
pNF

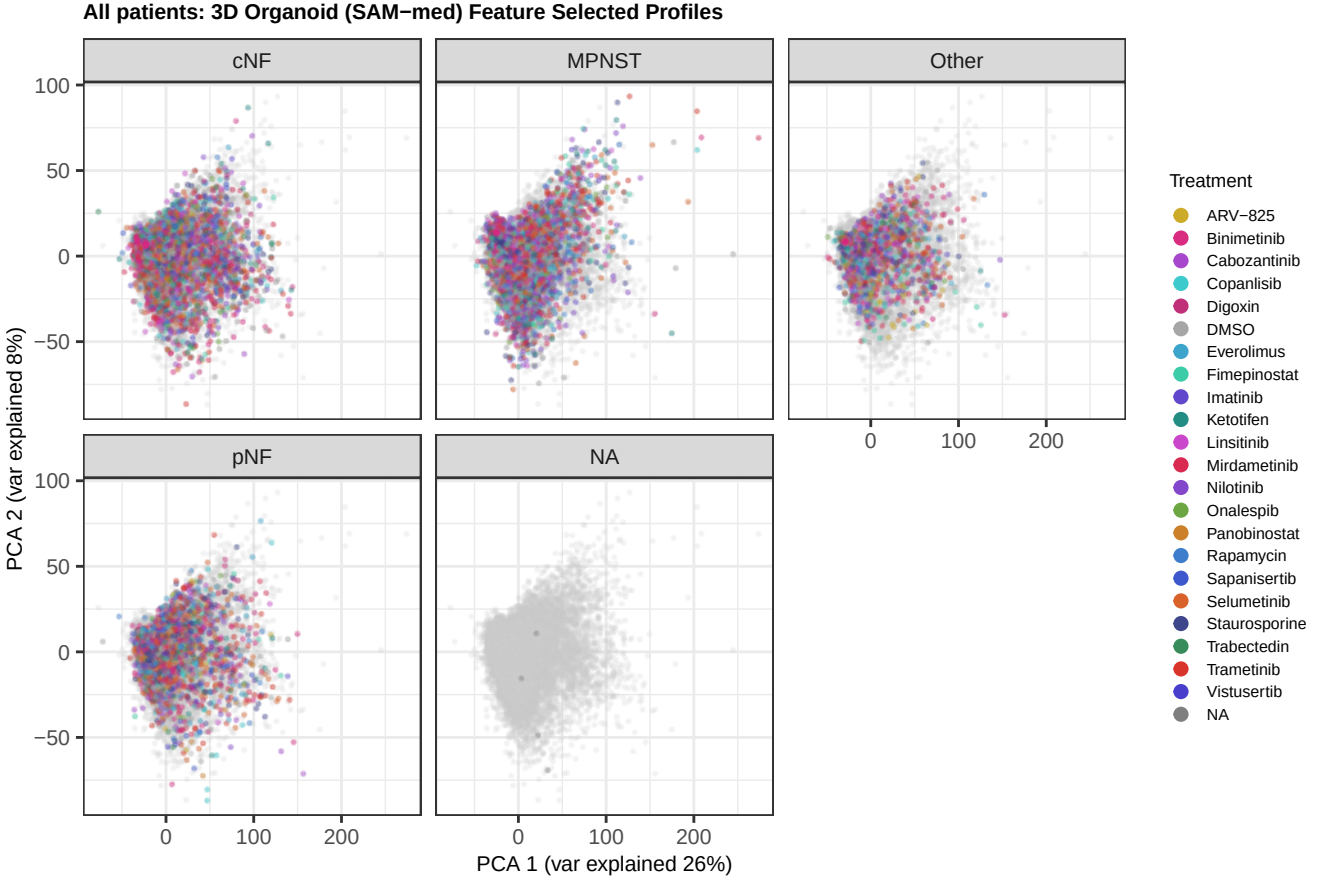

All patients: 3D Organoid (ZEDProfiler) Consensus Profiles
PCA 2 (var explained 11%)
PCA 1 (var explained 21%)
10
5
0
−5
−5
0
5
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA

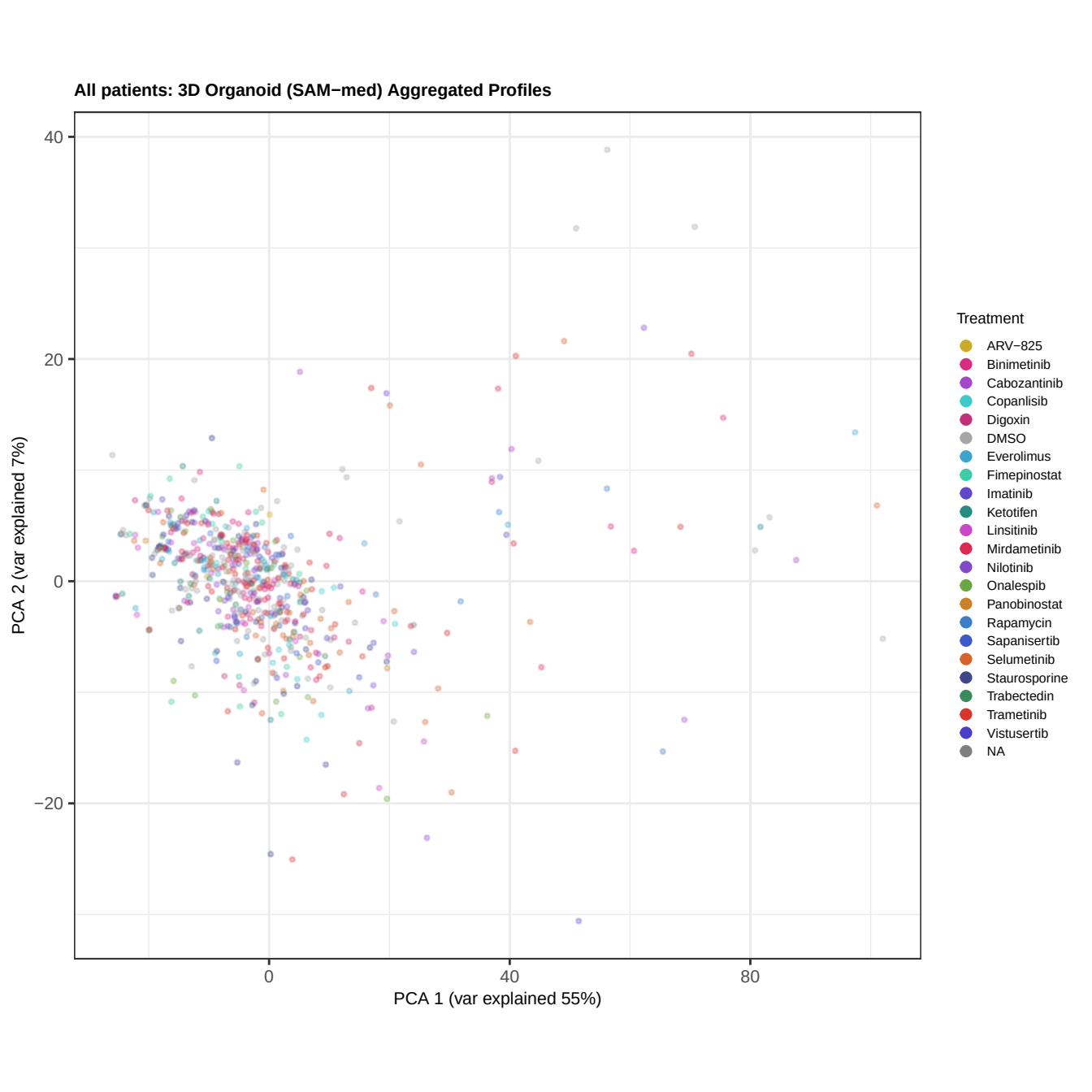

All patients: 3D Organoid (ZEDProfiler) Consensus Profiles
10
5
0
−5
PCA 2 (var explained 11%)
−5
0
5
PCA 1 (var explained 21%)
Tumor type
cNF
MPNST
Other
pNF

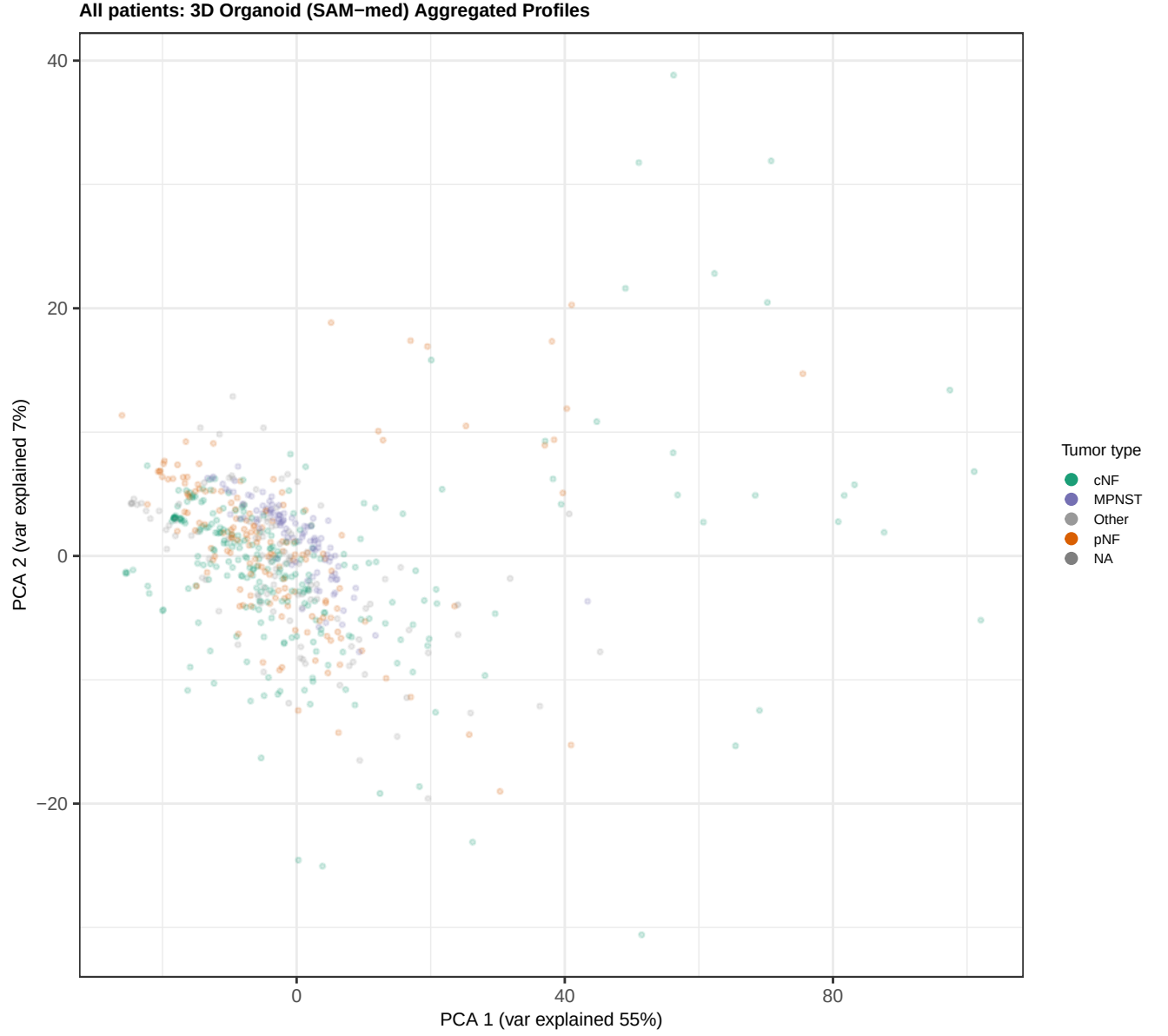

All patients: 3D Organoid (SAM−med) Feature Selected Profiles
100
50
0
−50
PCA 2 (var explained 8%)
0
100
200
PCA 1 (var explained 26%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA

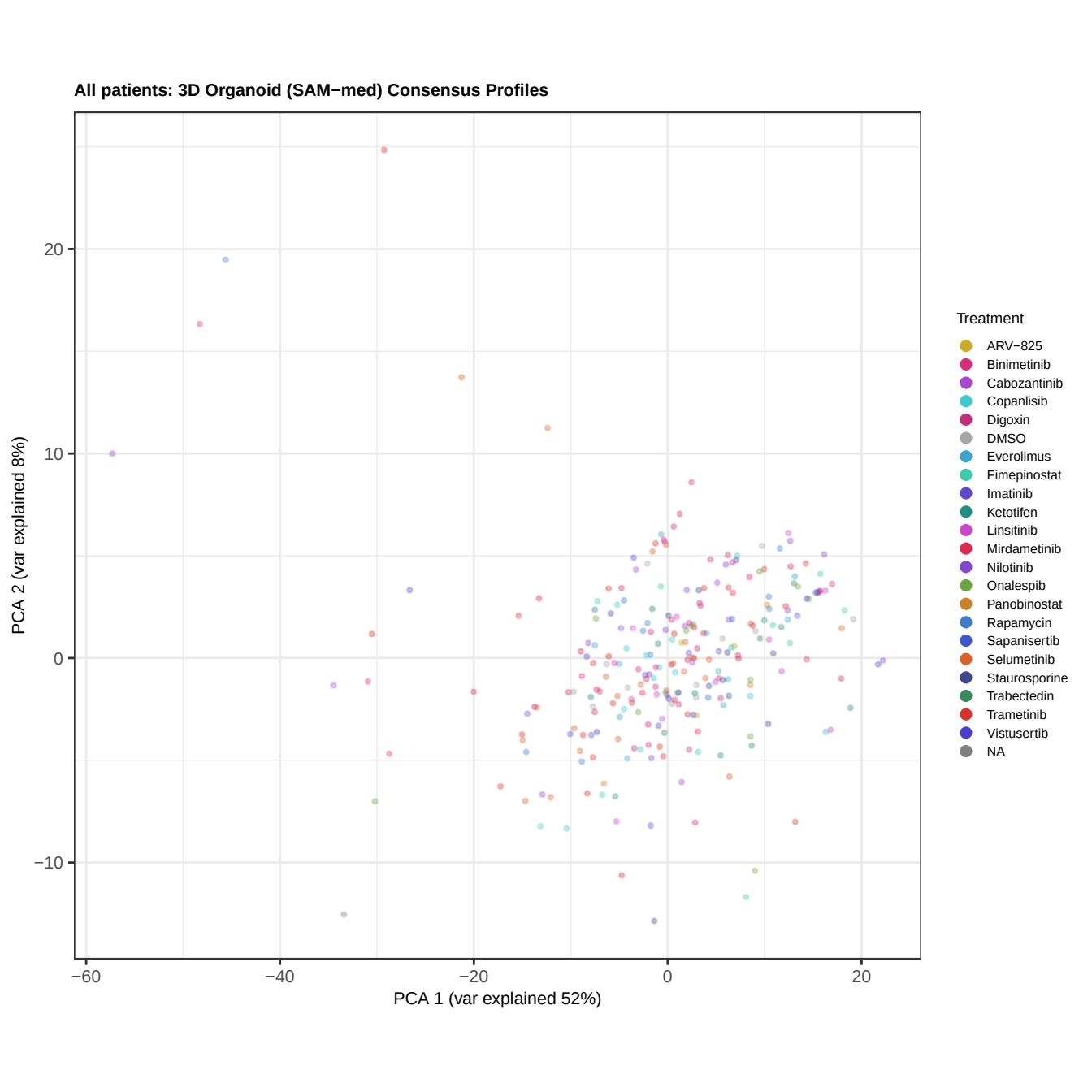

All patients: 3D Organoid (SAM−med) Feature Selected Profiles
100
50
0
−50
PCA 2 (var explained 8%)
0
100
200
PCA 1 (var explained 26%)
Tumor type
cNF
MPNST
Other
pNF
NA

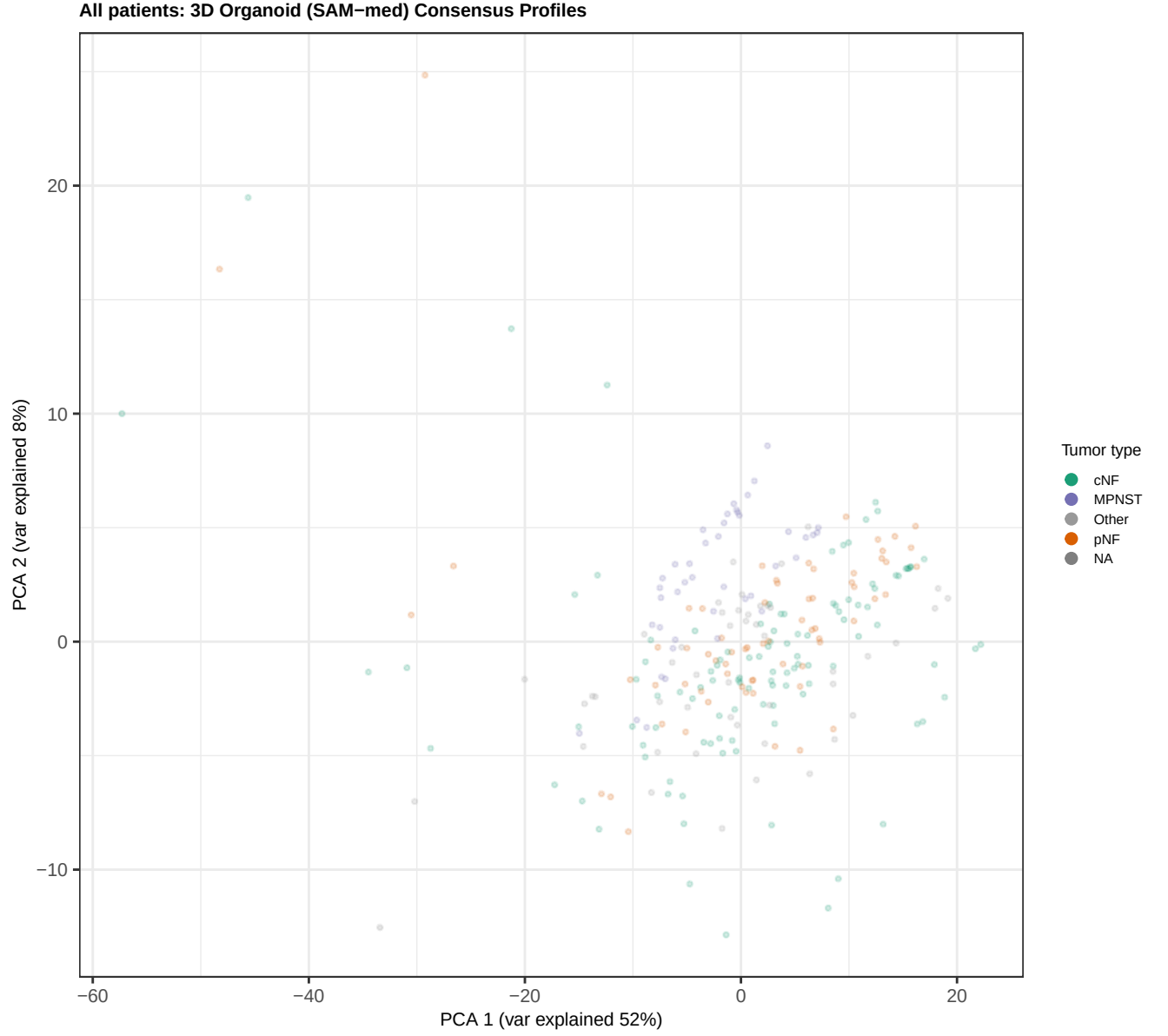

All patients: 3D Organoid (SAM−med) Feature Selected Profiles
NF0014_T1
NF0014_T2
NF0016_T1
NF0018_T6
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1
NA
100
50
0
−50
0
100
200
PCA 2 (var explained 8%)
PCA 1 (var explained 26%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA

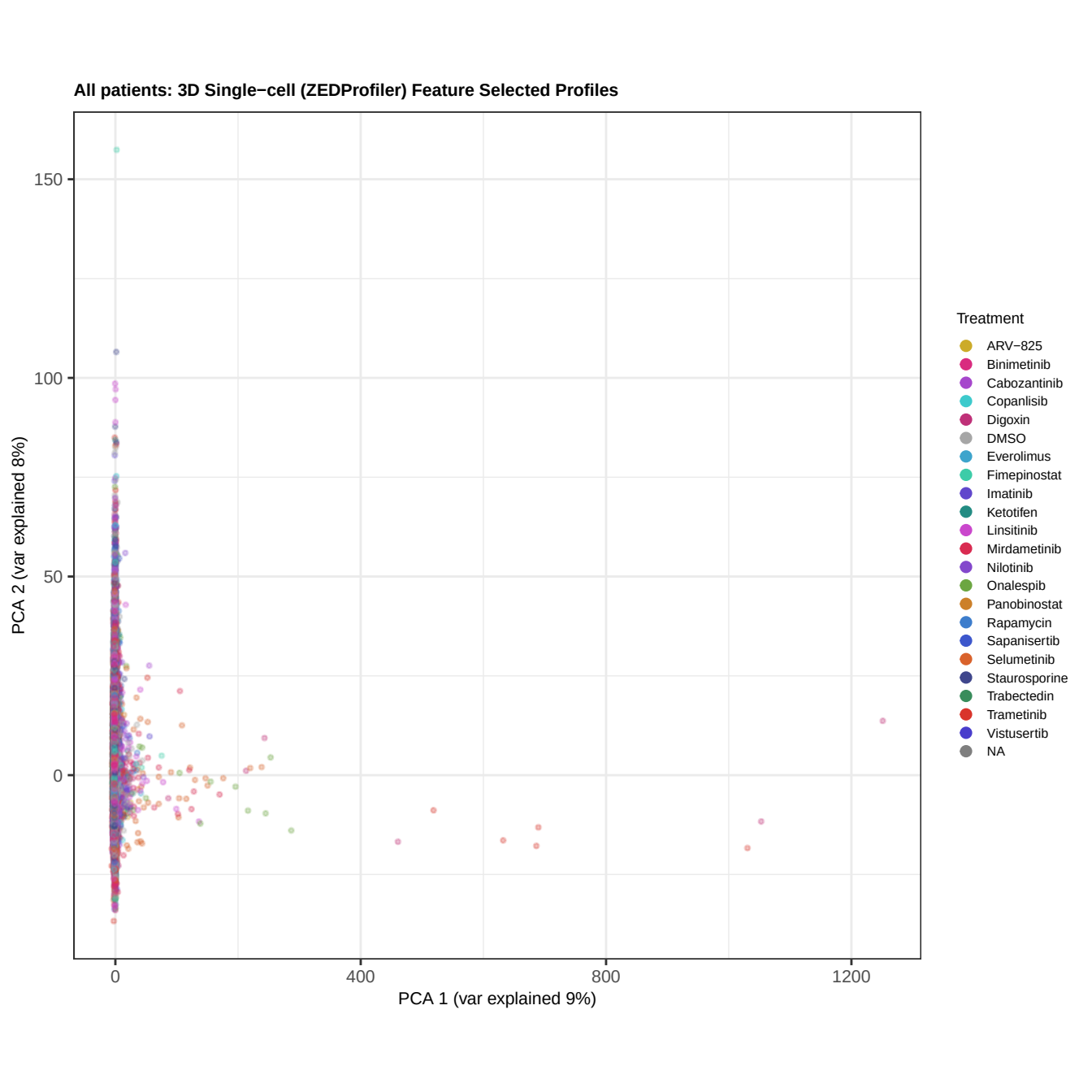

All patients: 3D Organoid (SAM−med) Feature Selected Profiles
cNF
MPNST
Other
pNF
NA
PCA 2 (var explained 8%)
PCA 1 (var explained 26%)
100
50
0
−50
0
100
200
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA

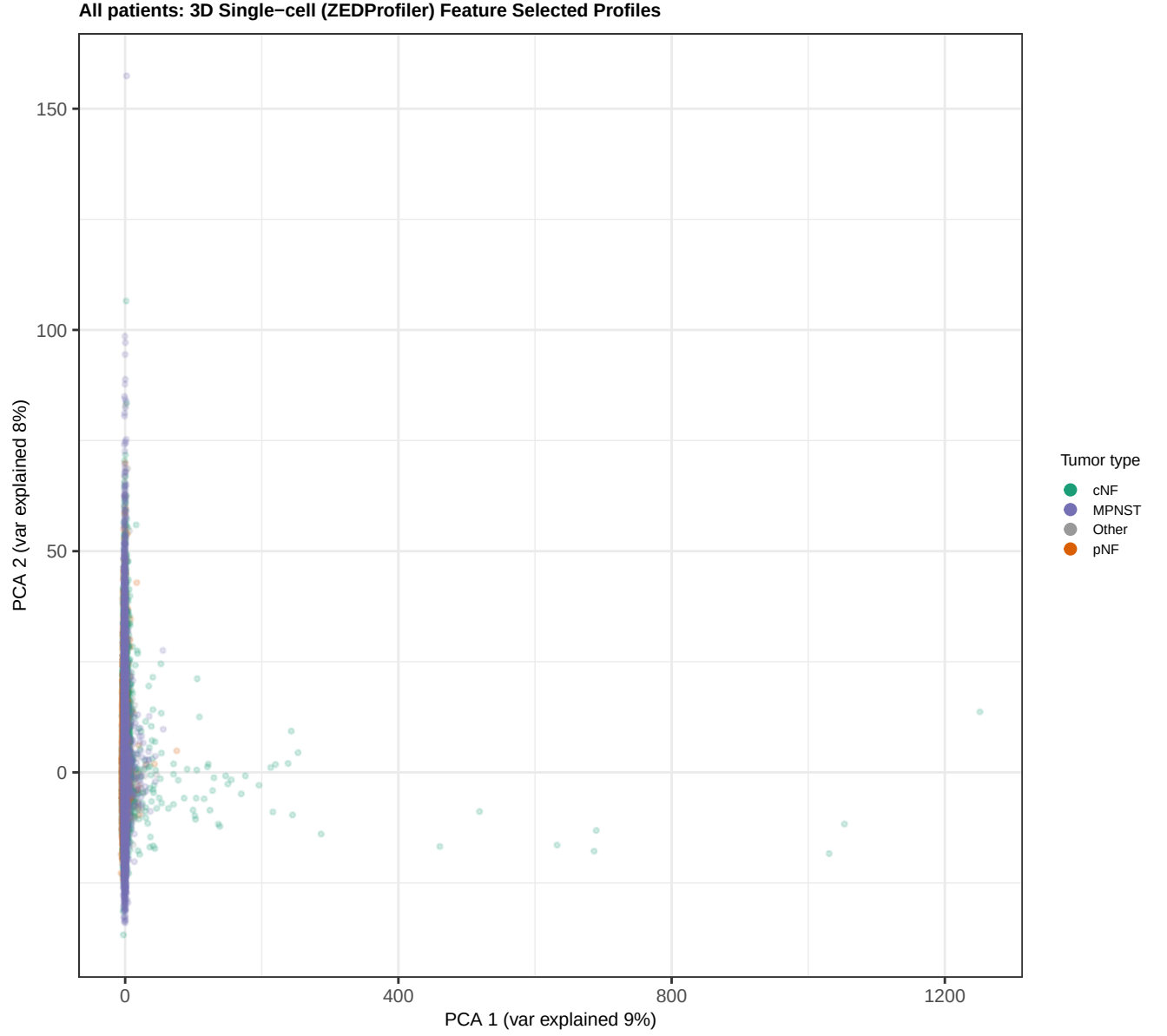

All patients: 3D Organoid (SAM−med) Aggregated Profiles
PCA 2 (var explained 7%)
PCA 1 (var explained 55%)
40
20
0
−20
0
40
80
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA

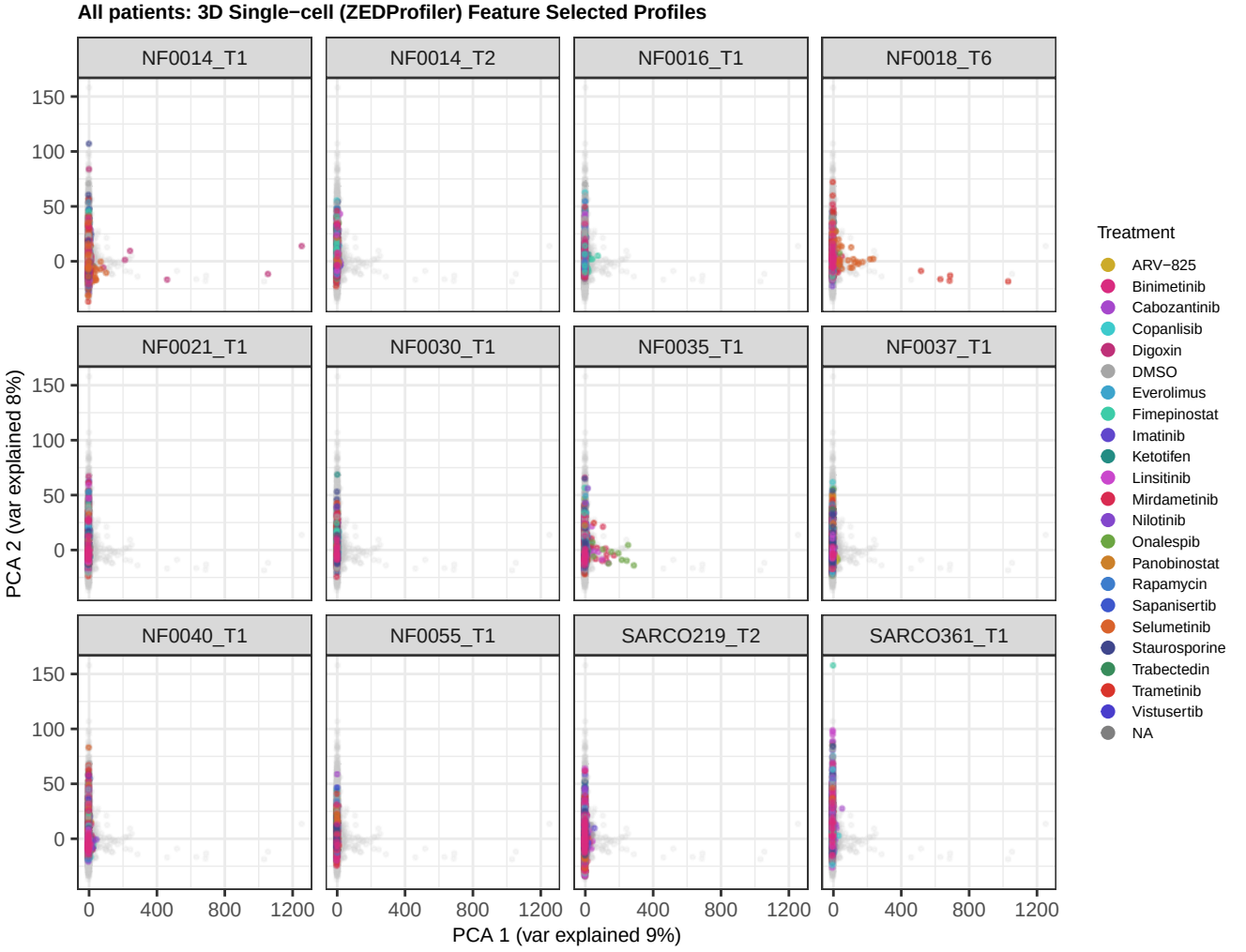

All patients: 3D Organoid (SAM−med) Aggregated Profiles
PCA 2 (var explained 7%)
40
20
0
−20
0
40
80
PCA 1 (var explained 55%)
Tumor type
cNF
MPNST
Other
pNF
NA

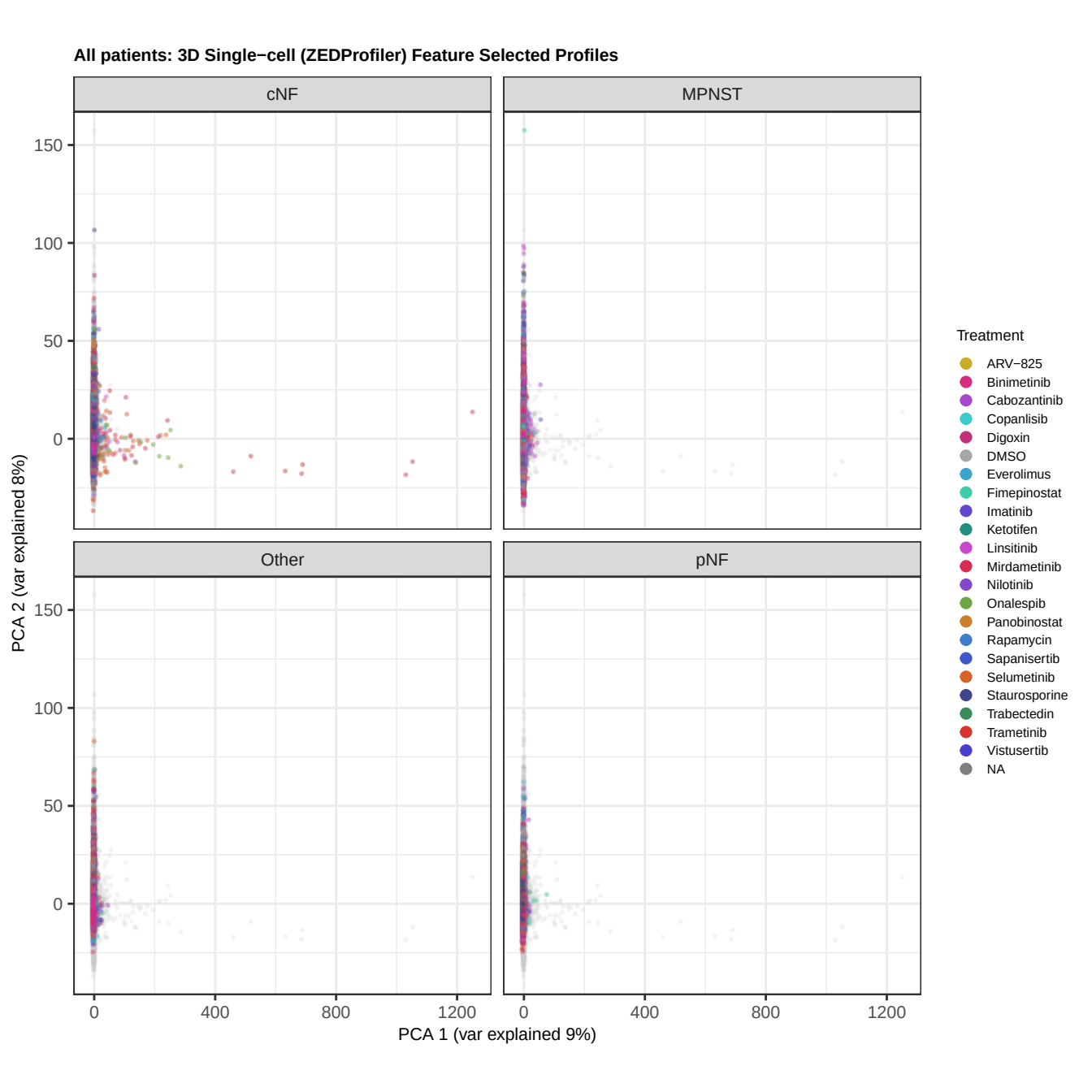

All patients: 3D Organoid (SAM−med) Consensus Profiles
PCA 2 (var explained 8%)
20
10
0
−10
−60
−40
−20
0
20
PCA 1 (var explained 52%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA

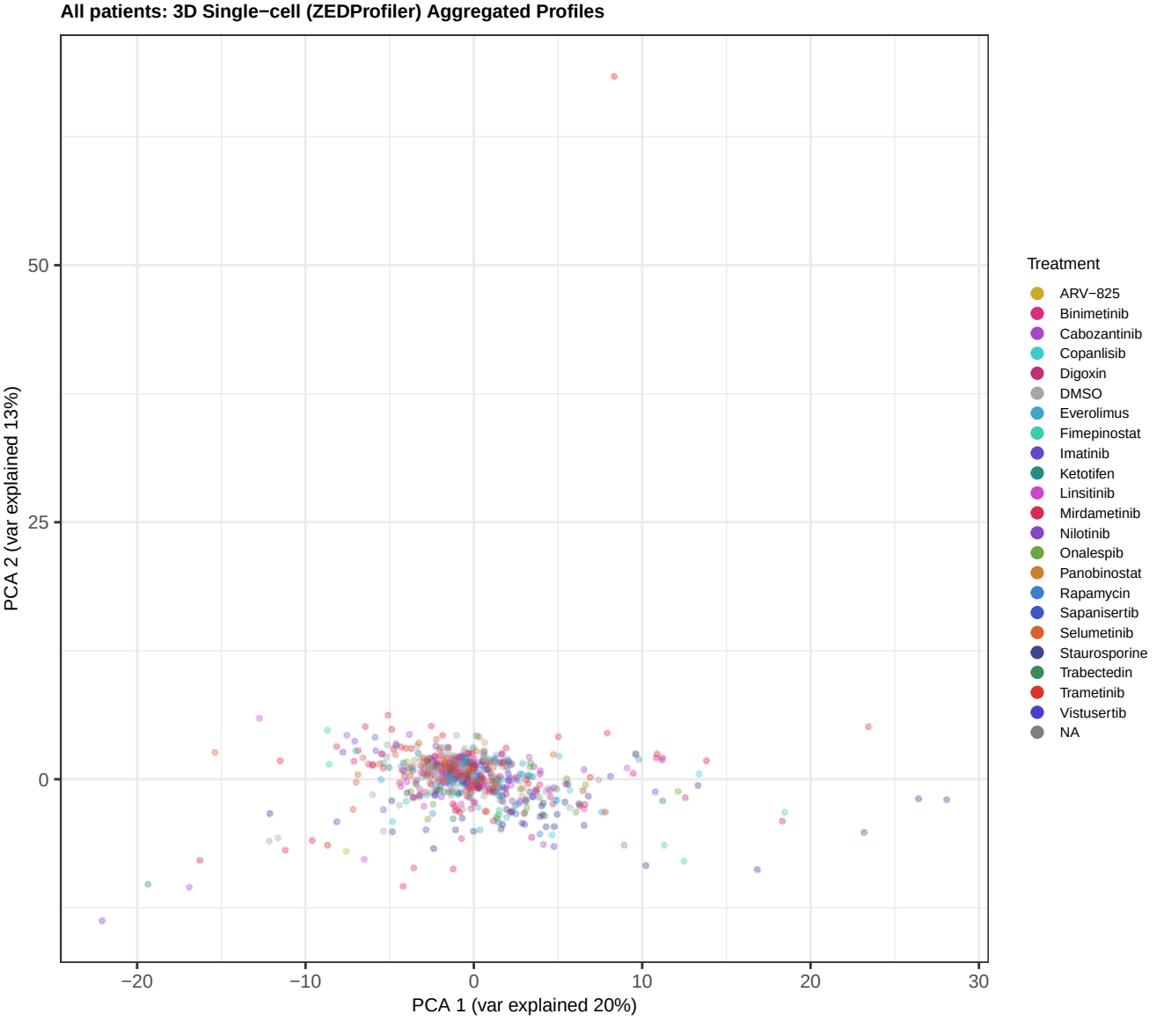

All patients: 3D Organoid (SAM−med) Consensus Profiles
PCA 2 (var explained 8%)
20
10
0
−10
−60
−40
−20
0
20
PCA 1 (var explained 52%)
Tumor type
cNF
MPNST
Other
pNF
NA

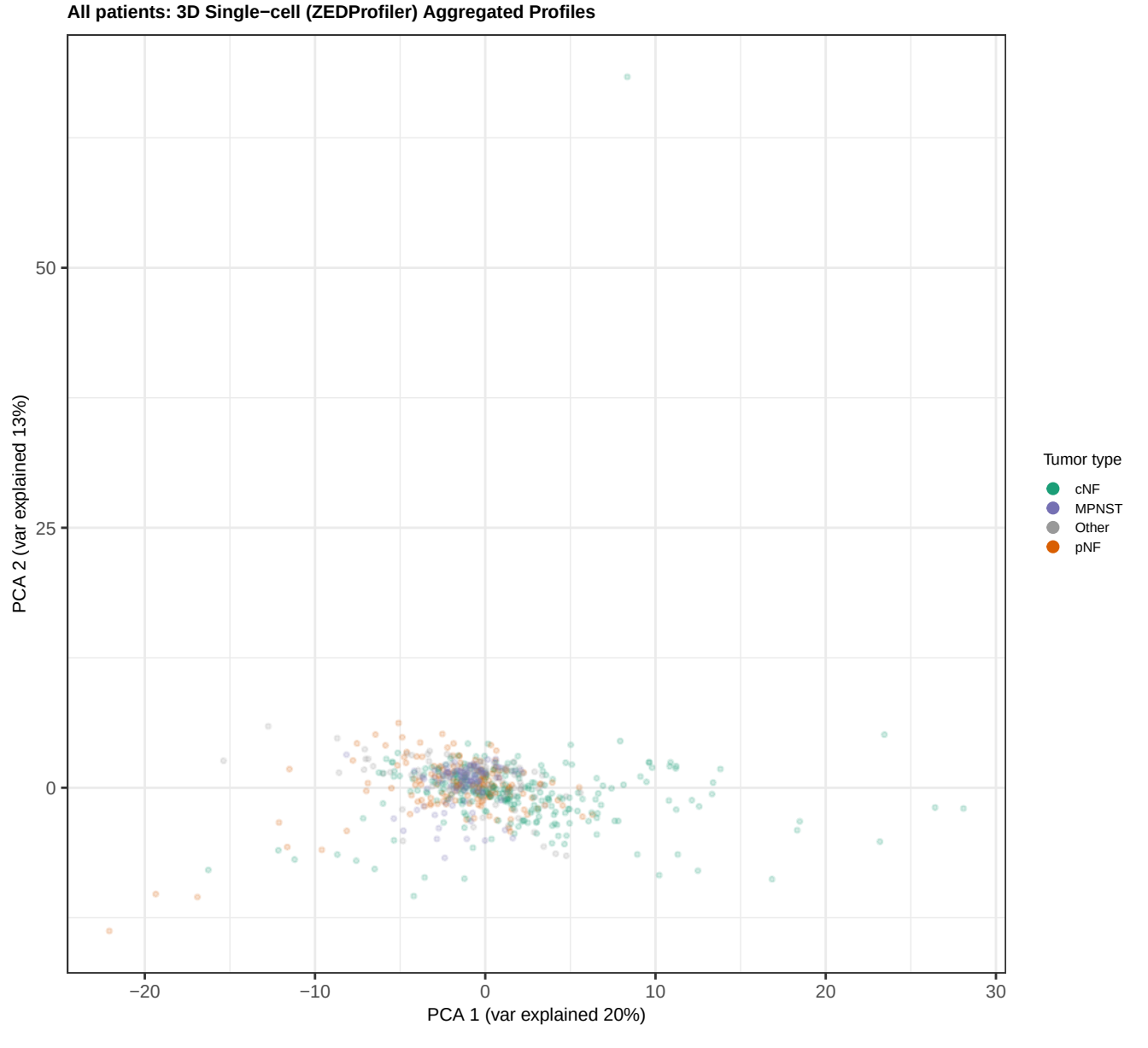

All patients: 3D Single−cell (ZEDProfiler) Feature Selected Profiles
PCA 2 (var explained 8%)
150
100
50
0
0
400
800
1200
PCA 1 (var explained 9%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA

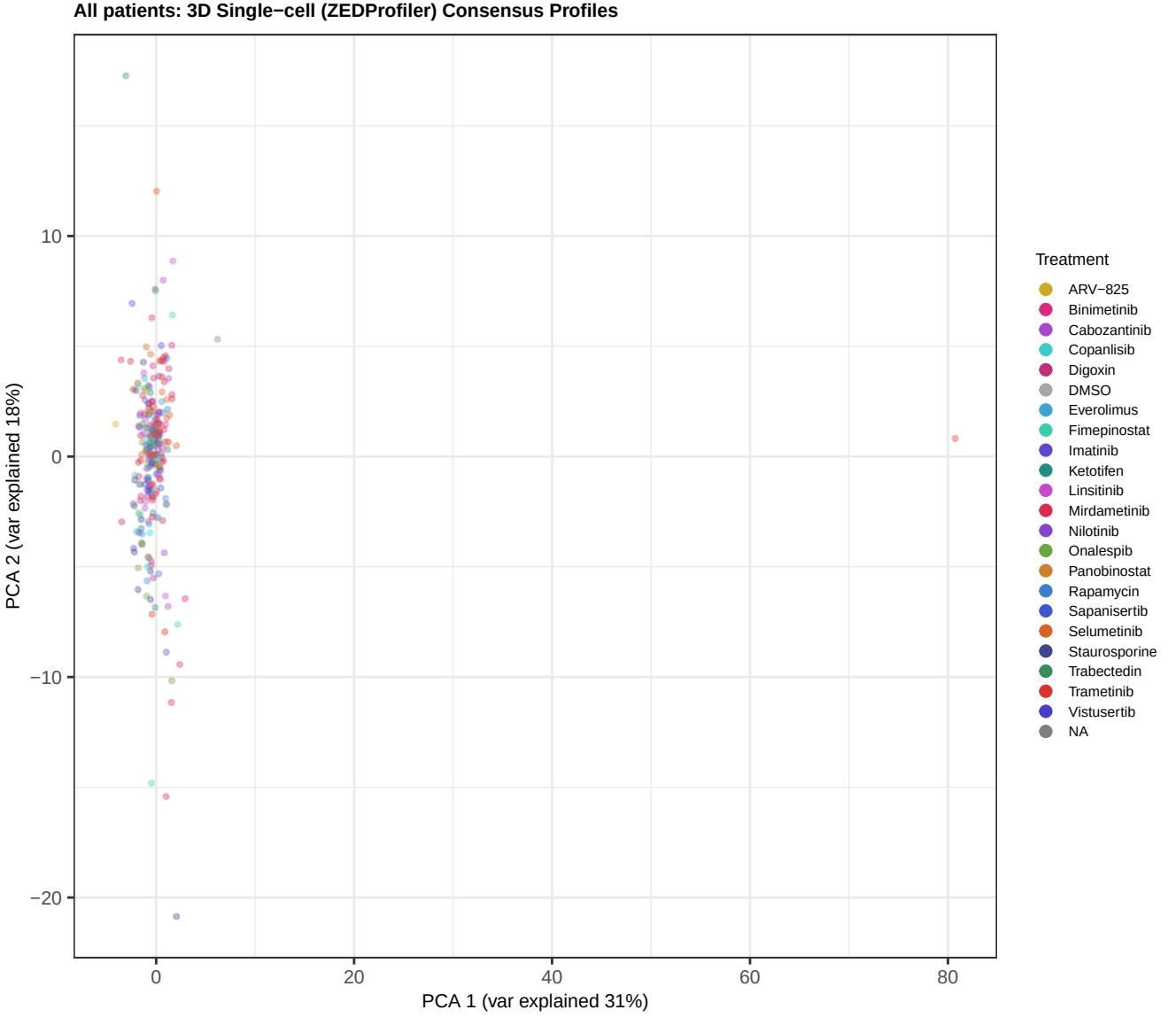

All patients: 3D Single−cell (ZEDProfiler) Feature Selected Profiles
150
100
50
0
PCA 2 (var explained 8%)
0
400
800
1200
PCA 1 (var explained 9%)
Tumor type
cNF
MPNST
Other
pNF

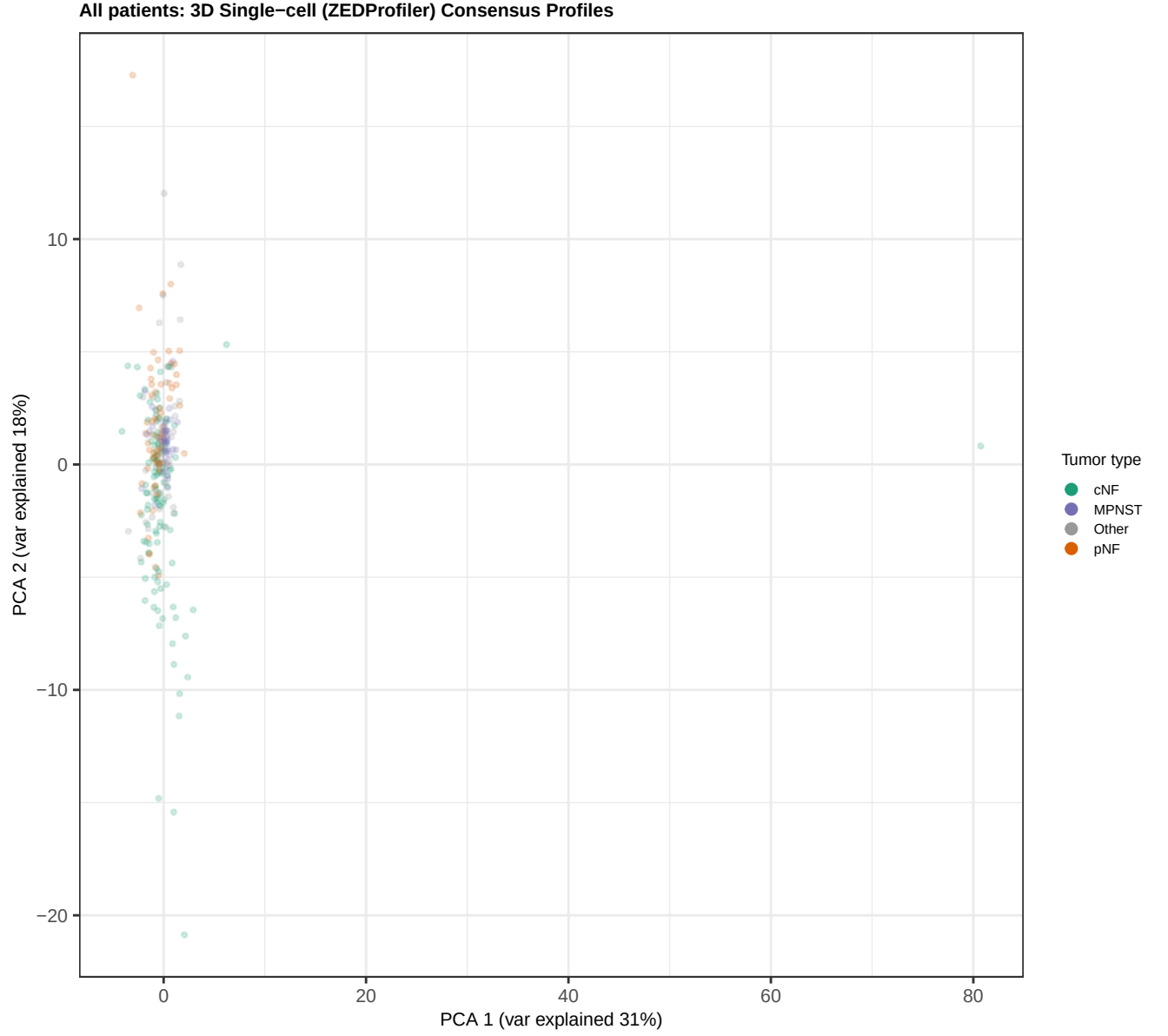

All patients: 3D Single−cell (ZEDProfiler) Feature Selected Profiles
NF0014_T1
NF0014_T2
NF0016_T1
NF0018_T6
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1
150
100
50
0
0
400
800
1200
PCA 2 (var explained 8%)
PCA 1 (var explained 9%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA

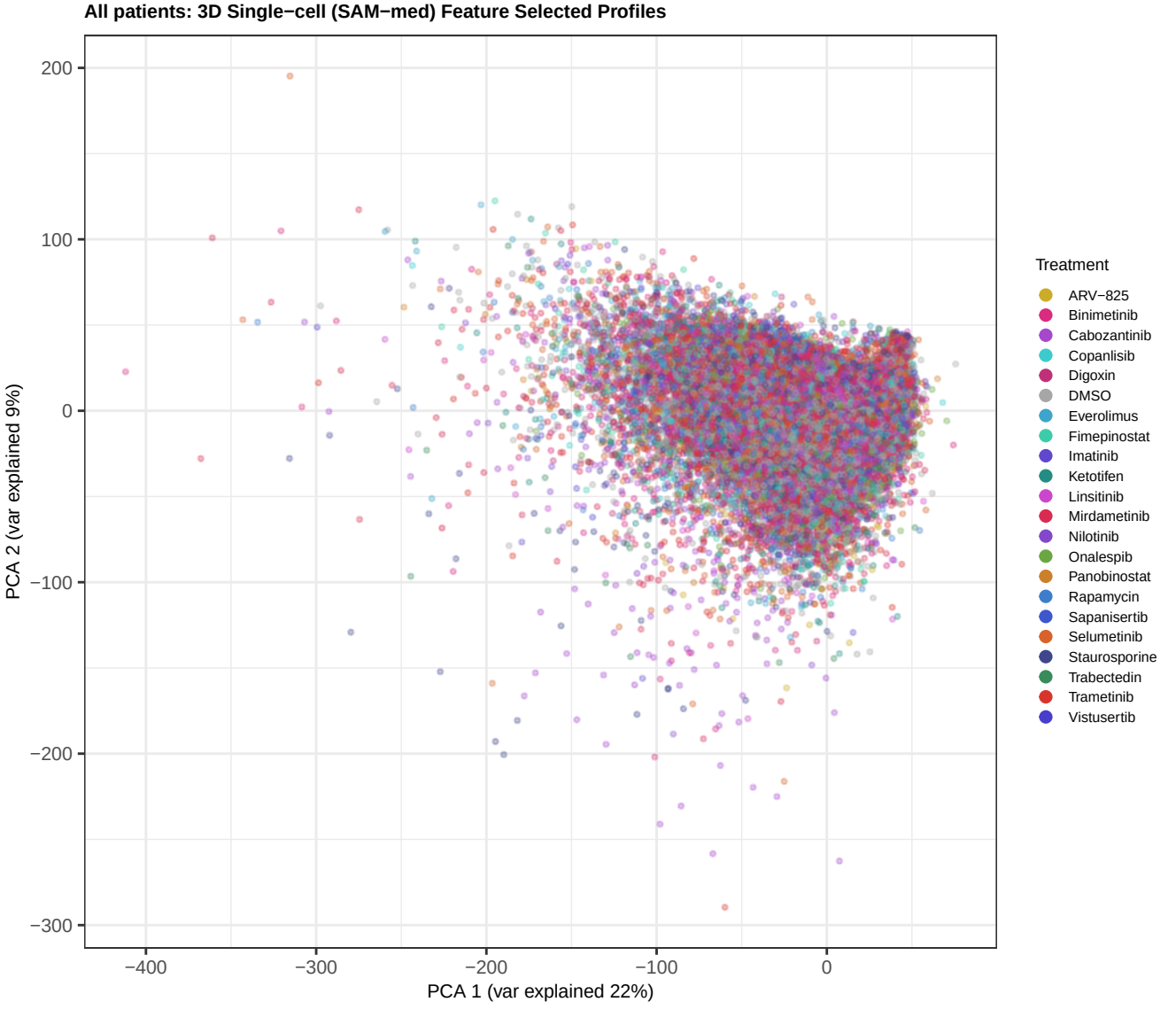

All patients: 3D Single−cell (ZEDProfiler) Feature Selected Profiles
cNF
MPNST
Other
pNF
PCA 2 (var explained 8%)
PCA 1 (var explained 9%)
150
100
50
0
400
800
1200
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA

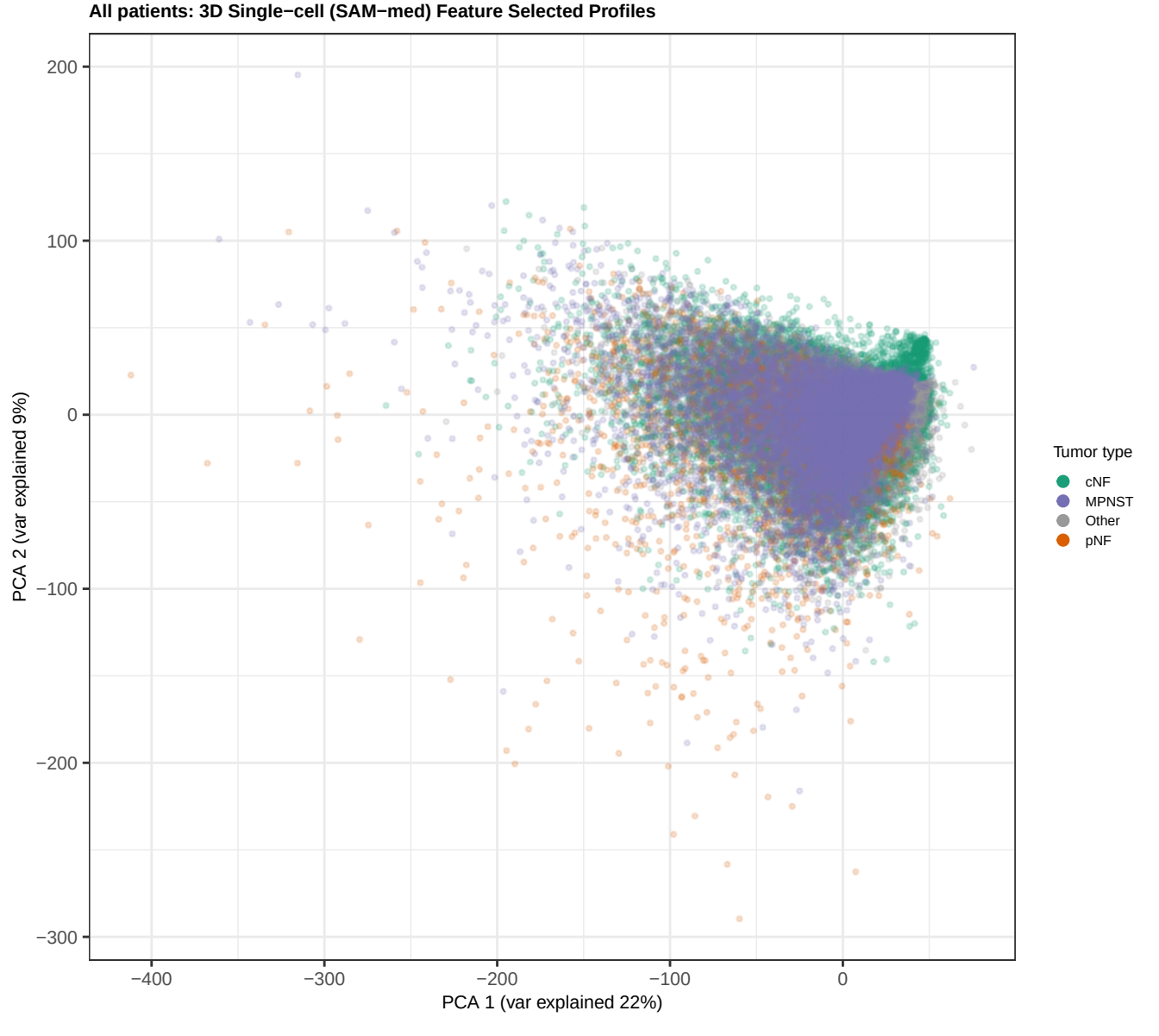

All patients: 3D Single−cell (ZEDProfiler) Aggregated Profiles
PCA 2 (var explained 13%)
50
25
0
−20
−10
0
10
20
30
PCA 1 (var explained 20%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA

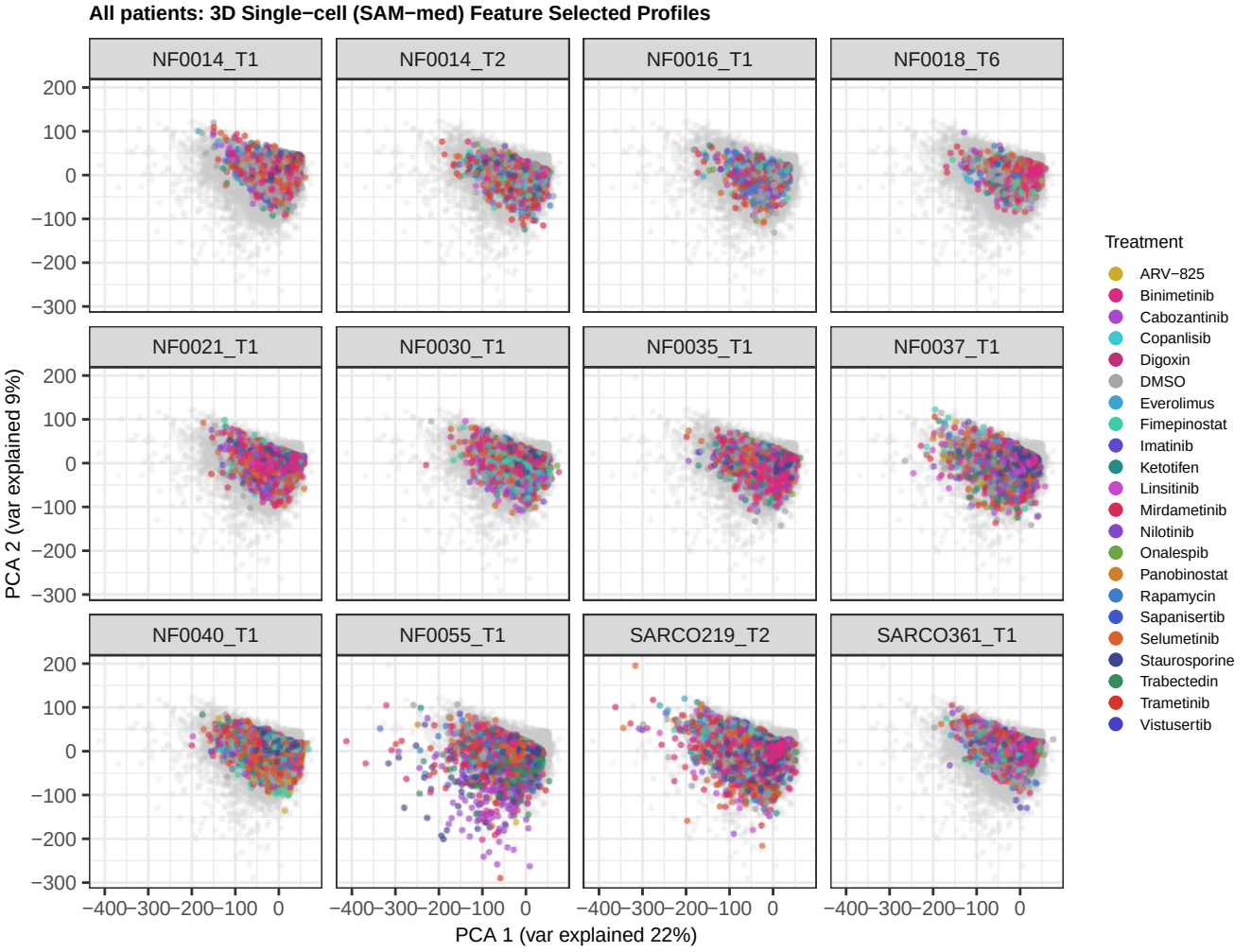

All patients: 3D Single−cell (ZEDProfiler) Aggregated Profiles
PCA 2 (var explained 13%)
50
25
0
−20
−10
0
10
20
30
PCA 1 (var explained 20%)
Tumor type
cNF
MPNST
Other
pNF

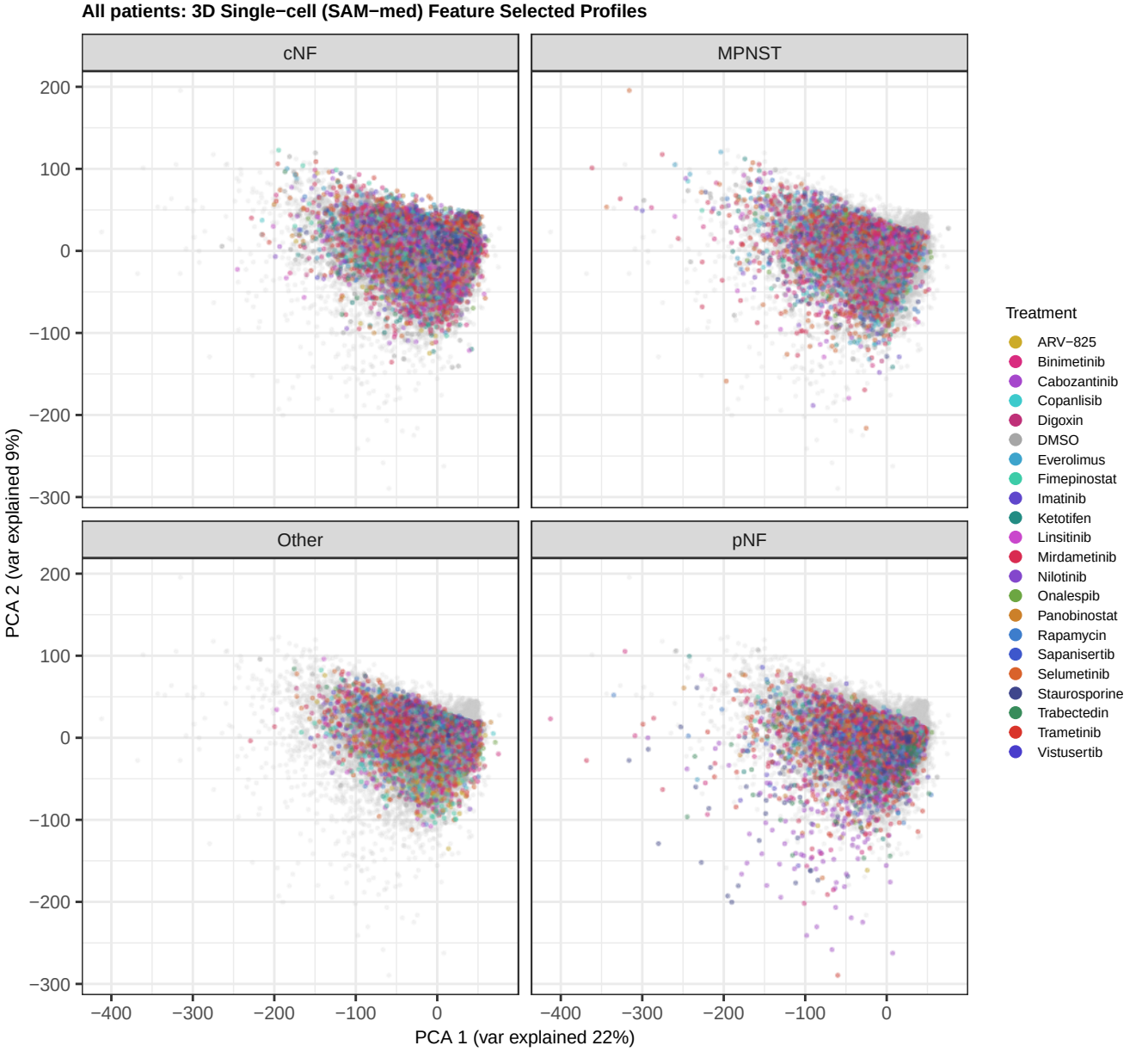

All patients: 3D Single−cell (ZEDProfiler) Consensus Profiles
PCA 2 (var explained 18%)
10
0
−10
−20
0
20
40
60
80
PCA 1 (var explained 31%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA

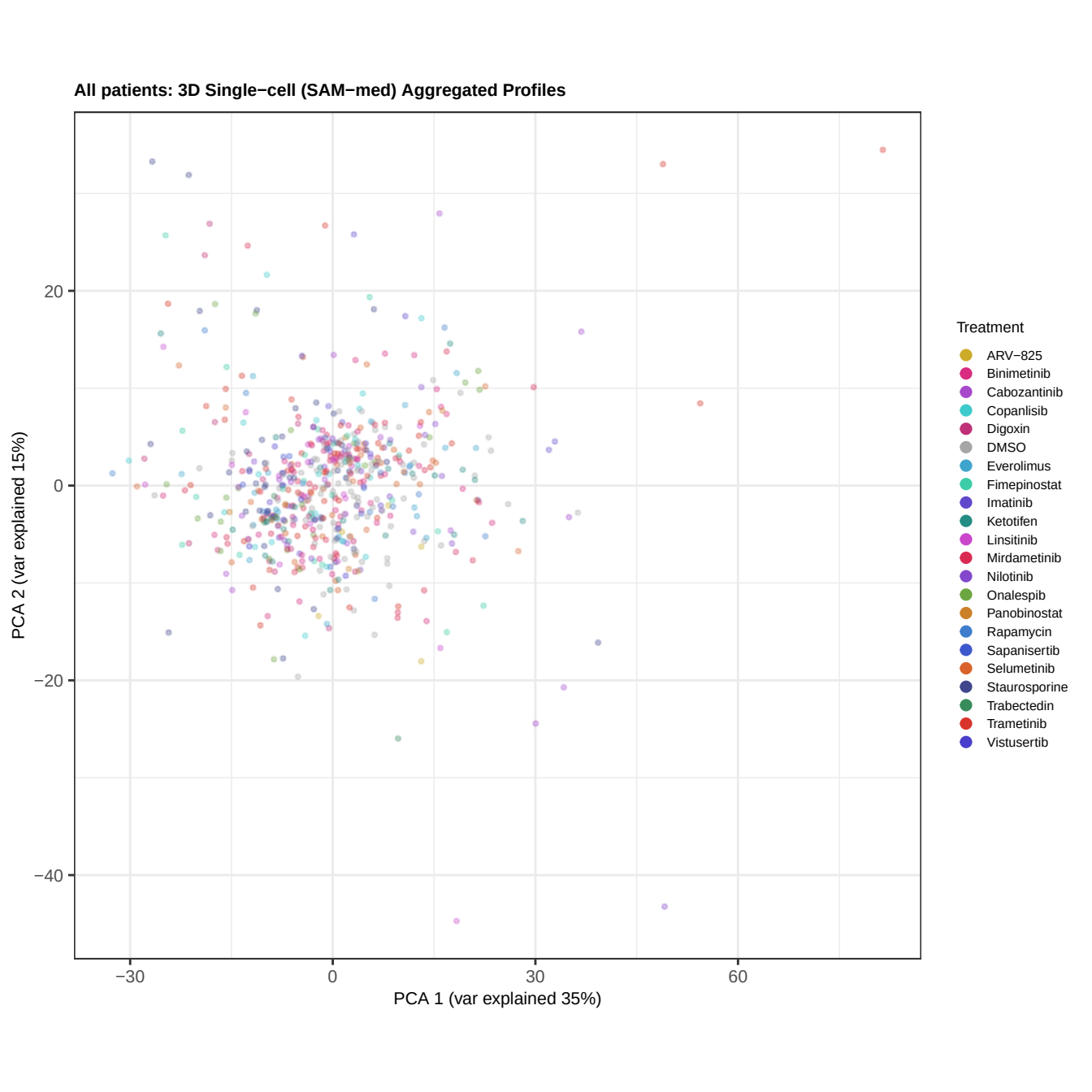

All patients: 3D Single−cell (ZEDProfiler) Consensus Profiles
PCA 2 (var explained 18%)
10
0
−10
−20
0
20
40
60
80
PCA 1 (var explained 31%)
Tumor type
cNF
MPNST
Other
pNF

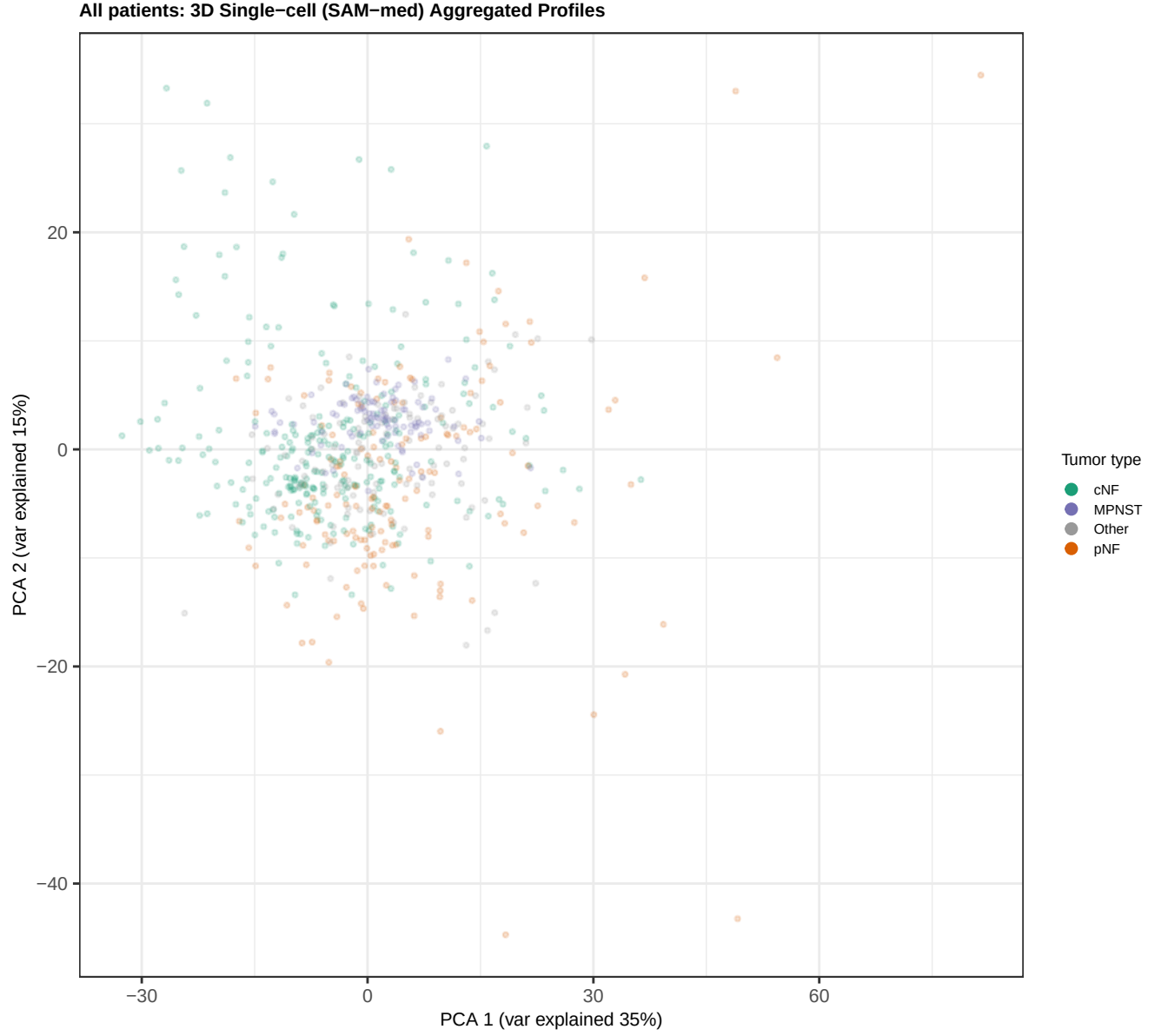

All patients: 3D Single−cell (SAM−med) Feature Selected Profiles
PCA 2 (var explained 9%)
200
100
0
−100
−200
−300
−400
−300
−200
−100
0
PCA 1 (var explained 22%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

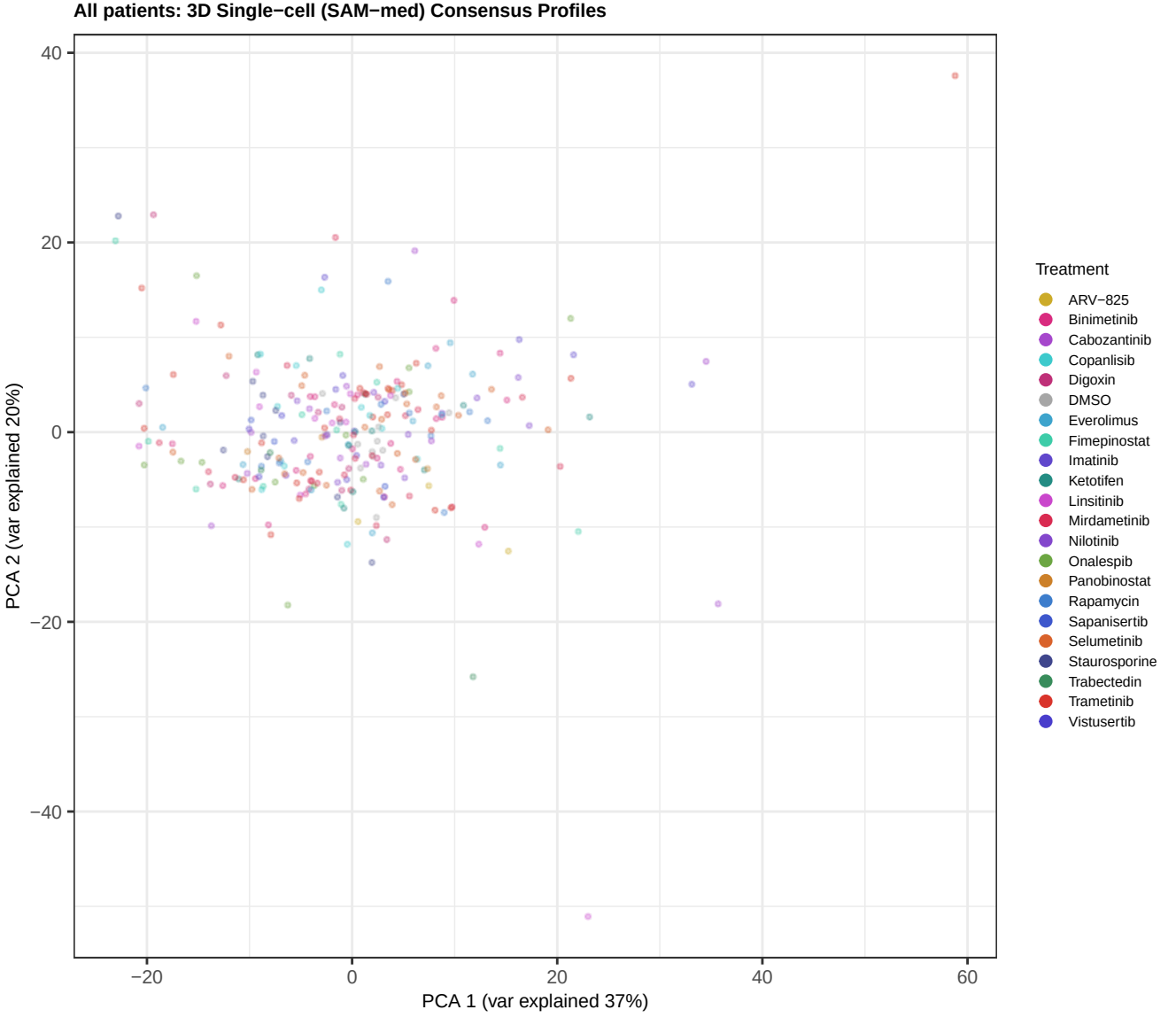

All patients: 3D Single−cell (SAM−med) Feature Selected Profiles
PCA 2 (var explained 9%)
200
100
0
−100
−200
−300
−400
−300
−200
−100
0
PCA 1 (var explained 22%)
Tumor type
cNF
MPNST
Other
pNF

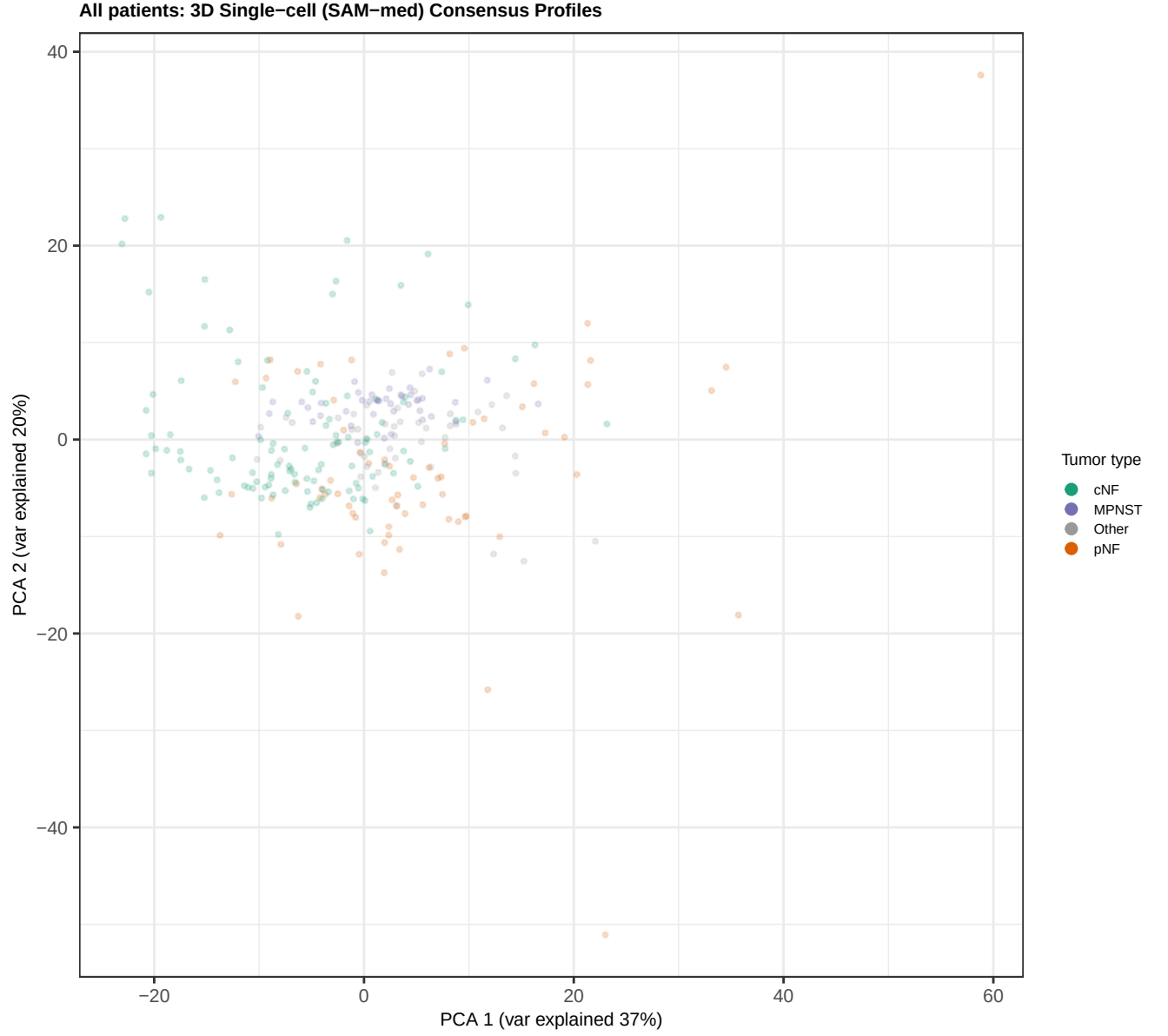

All patients: 3D Single−cell (SAM−med) Feature Selected Profiles
NF0014_T1
NF0014_T2
NF0016_T1
NF0018_T6
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1
PCA 2 (var explained 9%)
PCA 1 (var explained 22%)
200
100
0
−100
−200
−300
−400
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

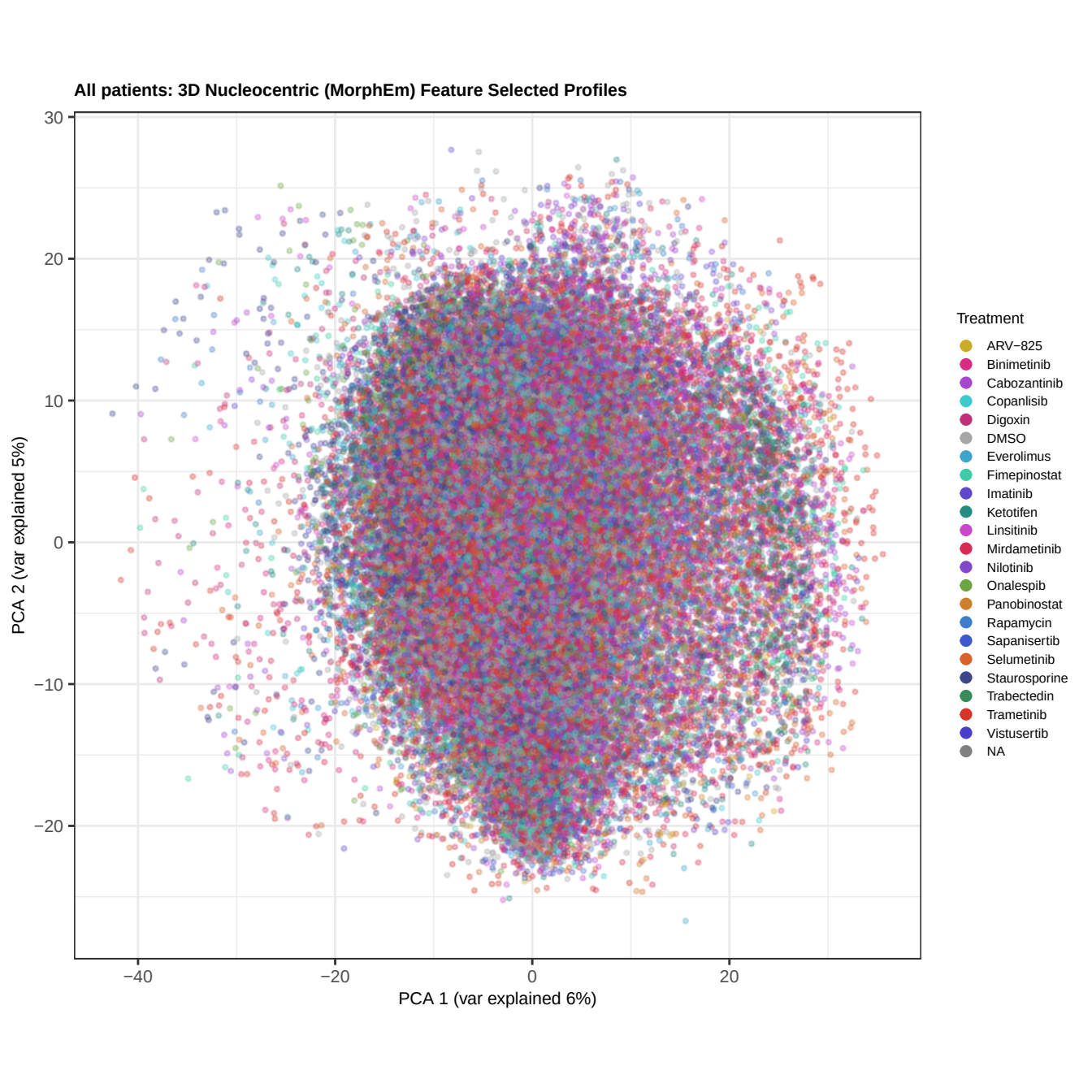

All patients: 3D Single−cell (SAM−med) Feature Selected Profiles
cNF
MPNST
Other
pNF
PCA 2 (var explained 9%)
PCA 1 (var explained 22%)
200
100
0
−100
−200
−300
−400
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

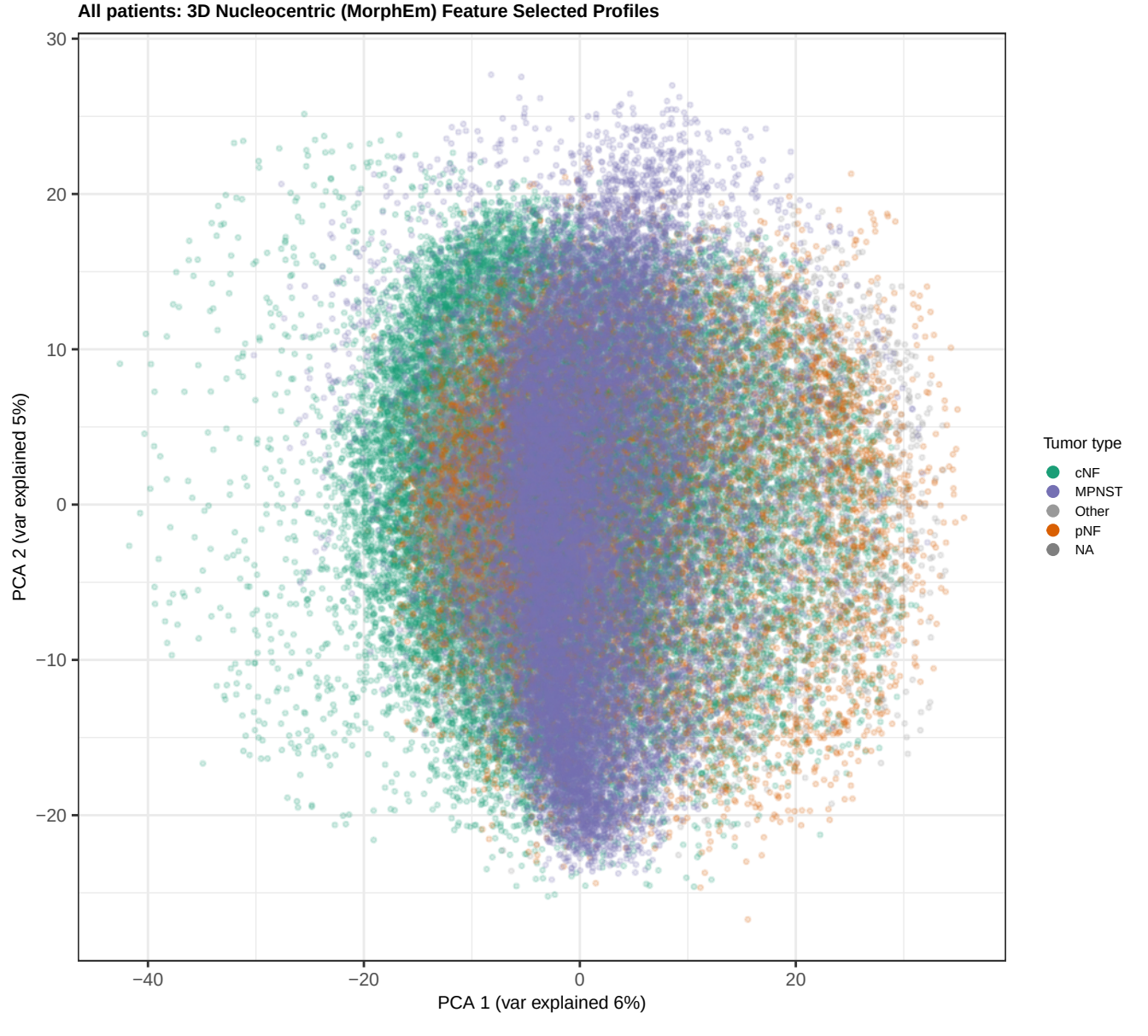

All patients: 3D Single−cell (SAM−med) Aggregated Profiles
PCA 2 (var explained 15%)
20
0
−20
−40
−30
0
30
60
PCA 1 (var explained 35%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

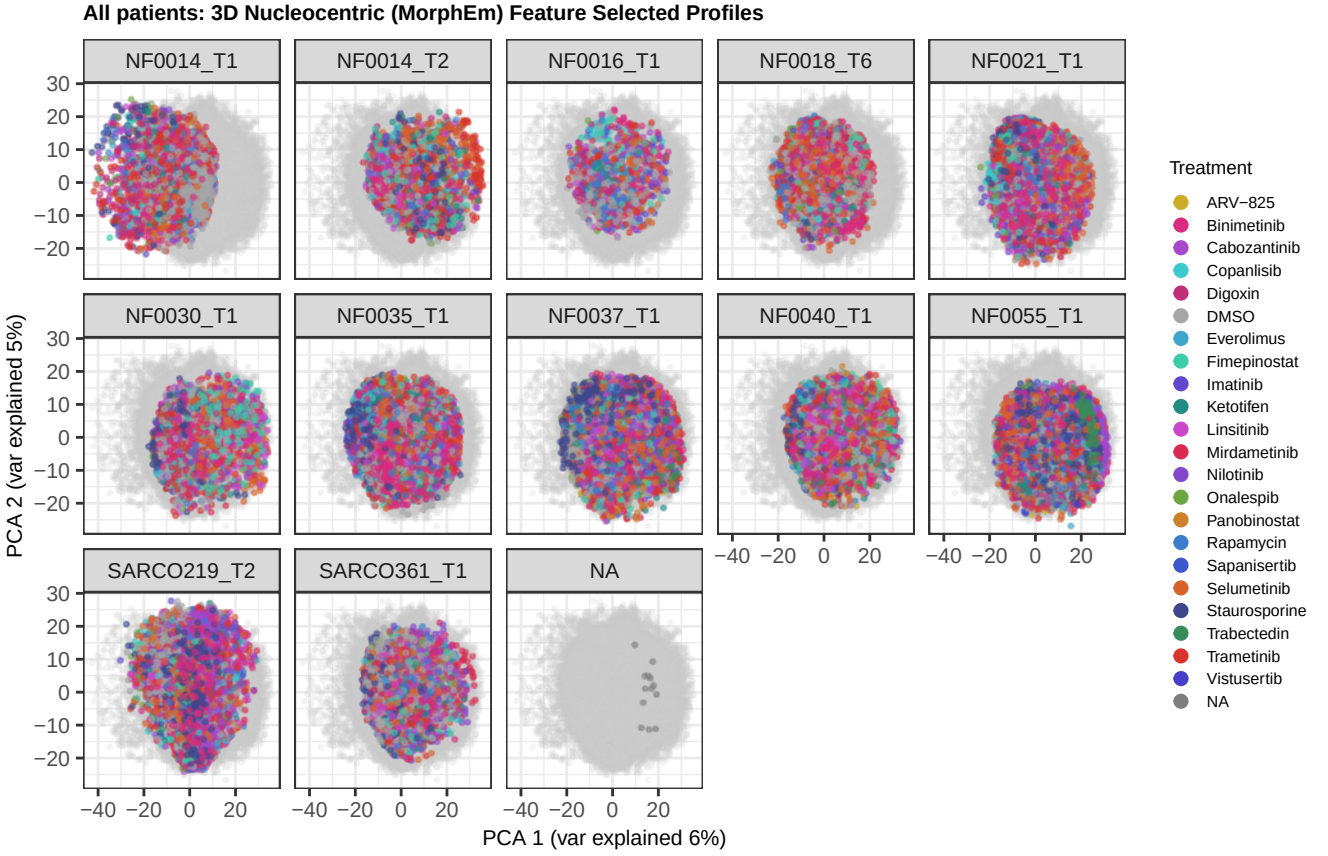

All patients: 3D Single−cell (SAM−med) Aggregated Profiles
PCA 2 (var explained 15%)
20
0
−20
−40
−30
0
30
60
PCA 1 (var explained 35%)
Tumor type
cNF
MPNST
Other
pNF

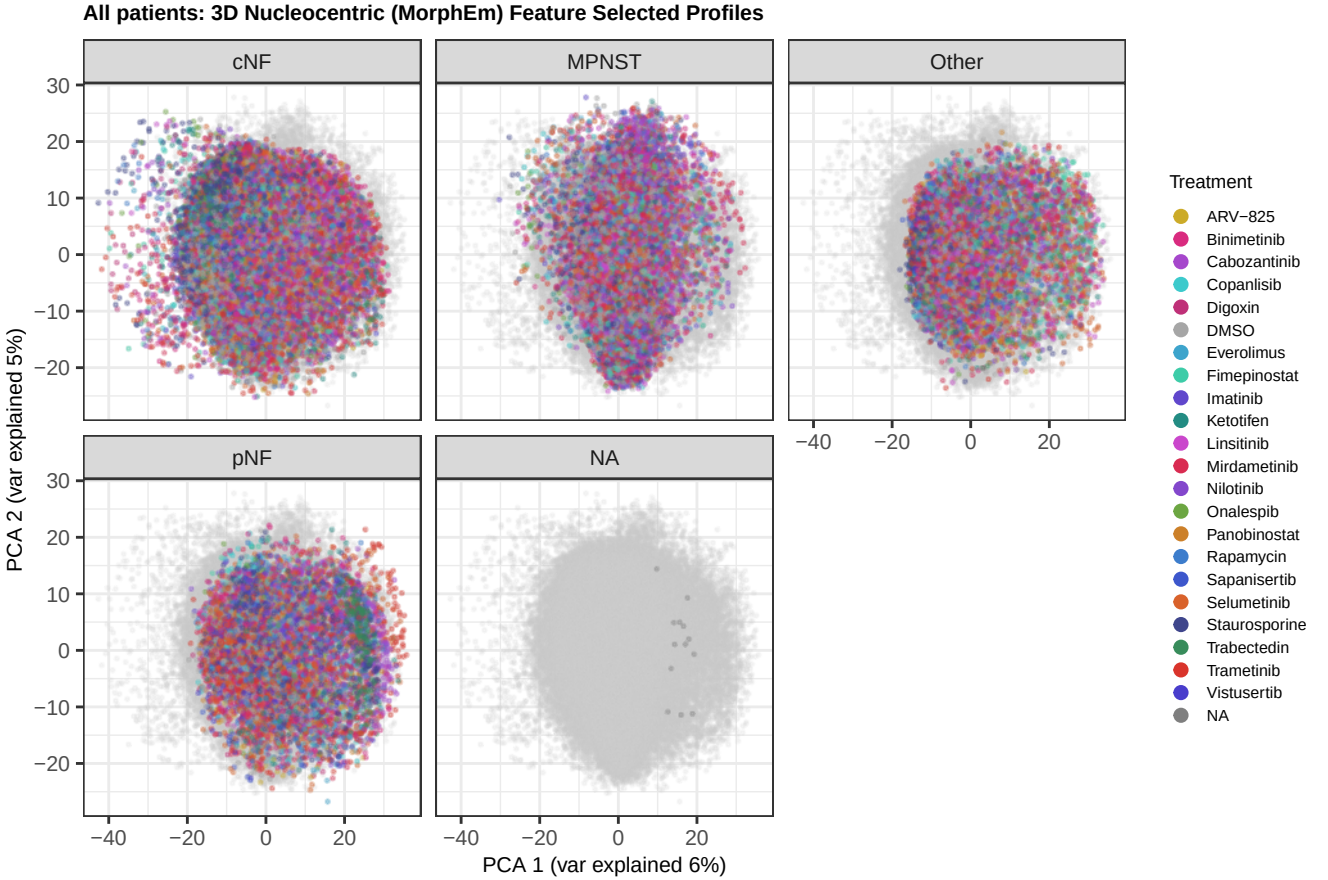

All patients: 3D Single−cell (SAM−med) Consensus Profiles
PCA 2 (var explained 20%)
PCA 1 (var explained 37%)
40
20
0
−20
−40
−20
0
20
40
60
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib

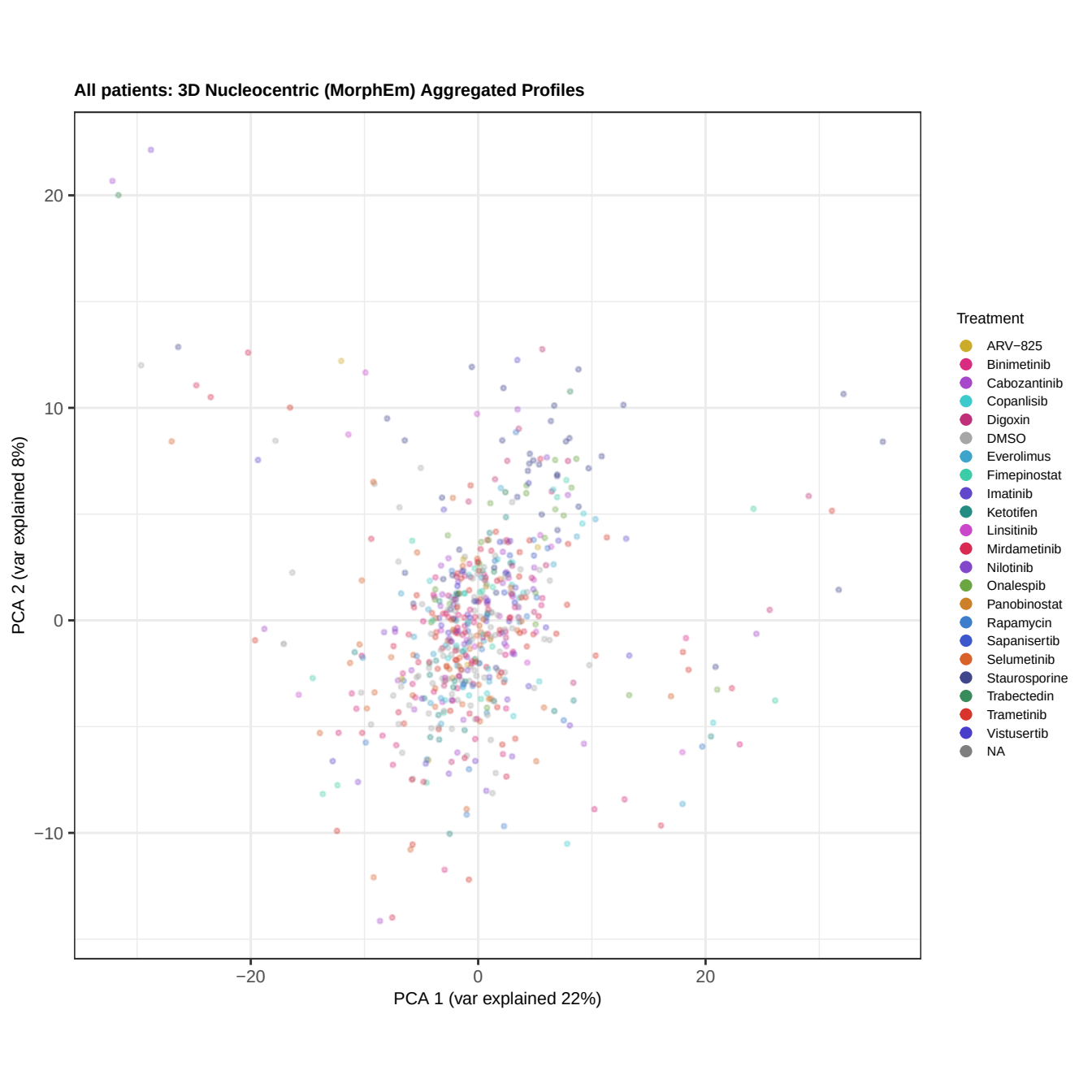

All patients: 3D Single−cell (SAM−med) Consensus Profiles
PCA 2 (var explained 20%)
PCA 1 (var explained 37%)
40
20
0
−20
−40
−20
0
20
40
60
Tumor type
cNF
MPNST
Other
pNF

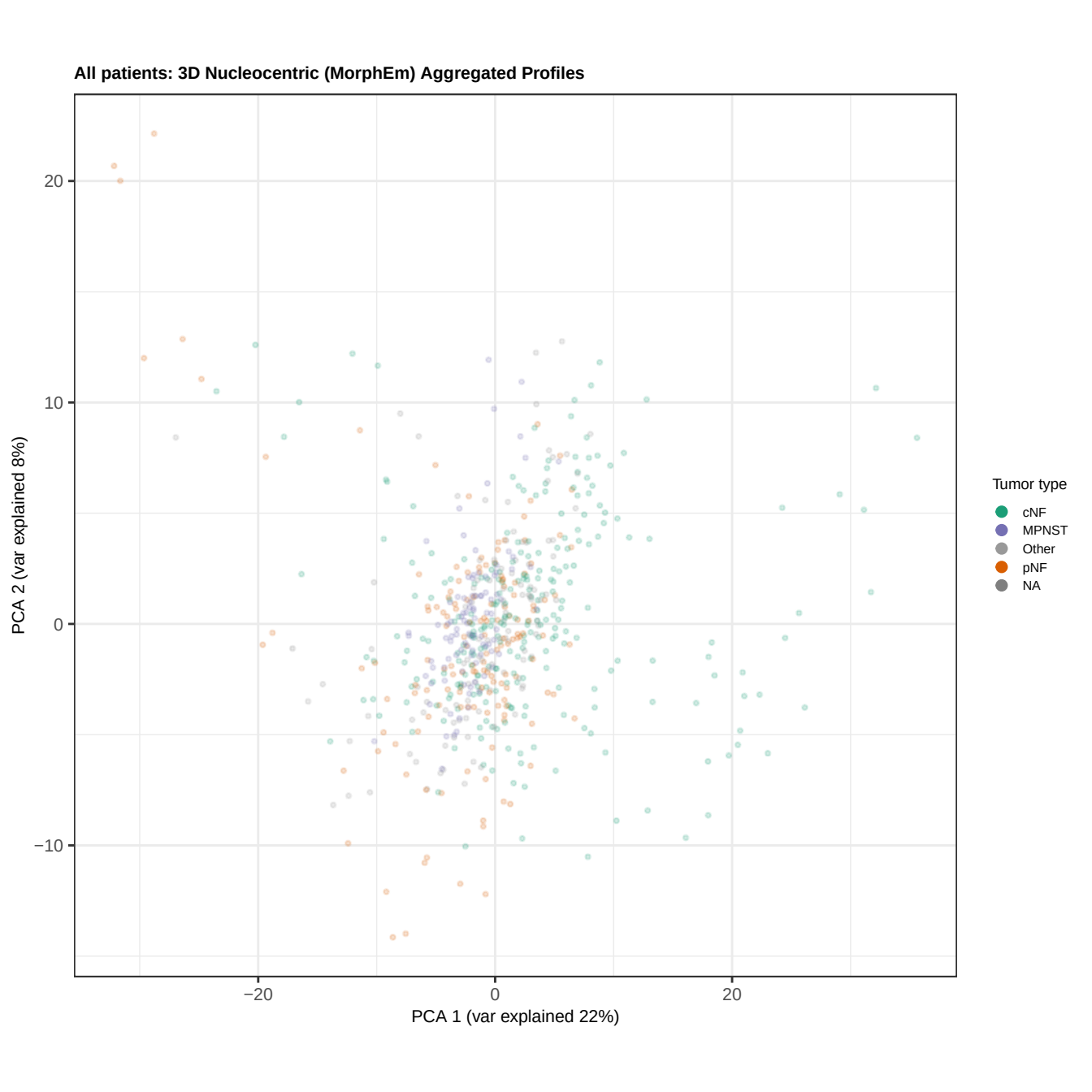

All patients: 3D Nucleocentric (MorphEm) Feature Selected Profiles
PCA 2 (var explained 5%)
PCA 1 (var explained 6%)
30
20
10
0
−10
−20
−40
−20
0
20
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA

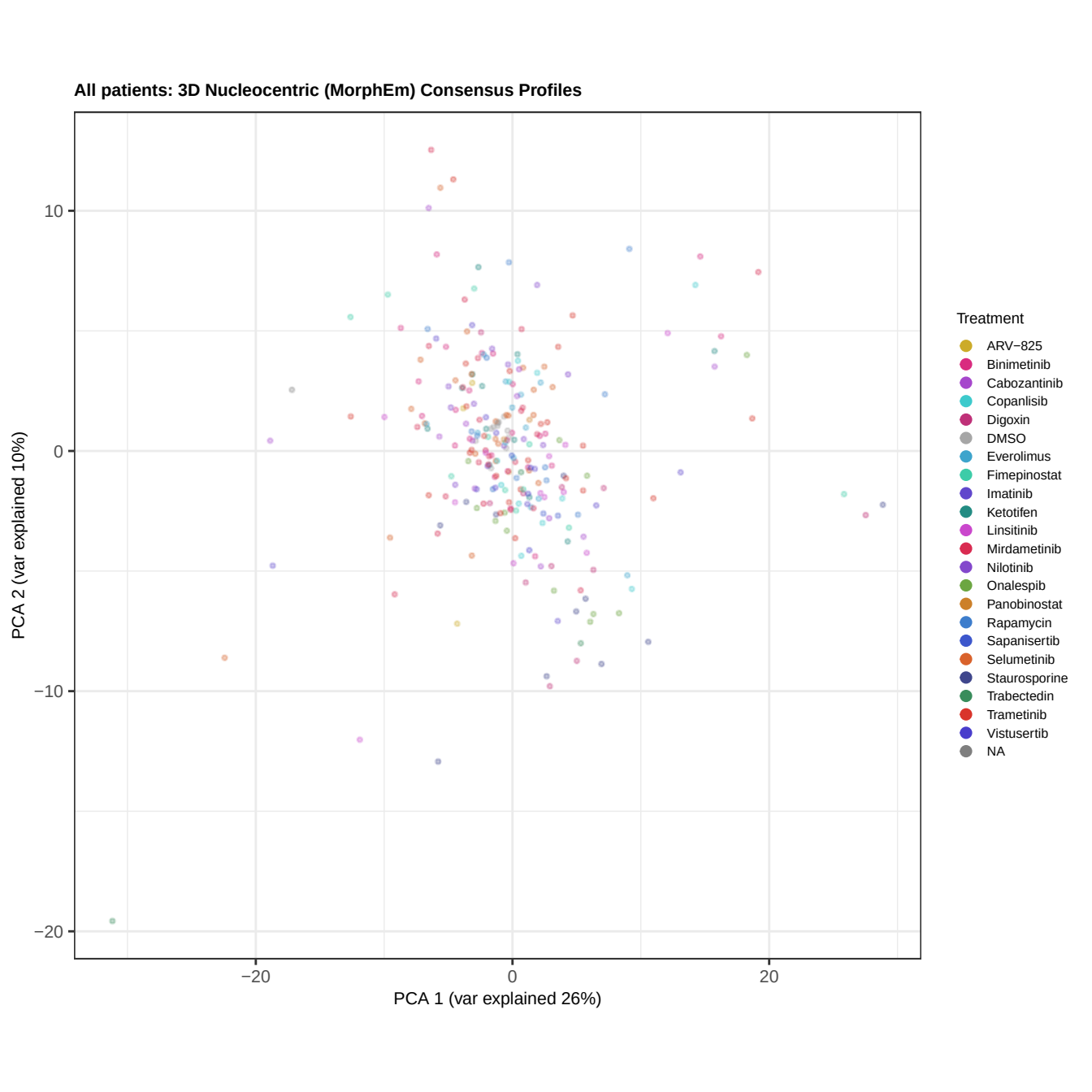

All patients: 3D Nucleocentric (MorphEm) Feature Selected Profiles
30
20
10
0
−10
−20
PCA 2 (var explained 5%)
−40
−20
0
20
PCA 1 (var explained 6%)
Tumor type
cNF
MPNST
Other
pNF
NA

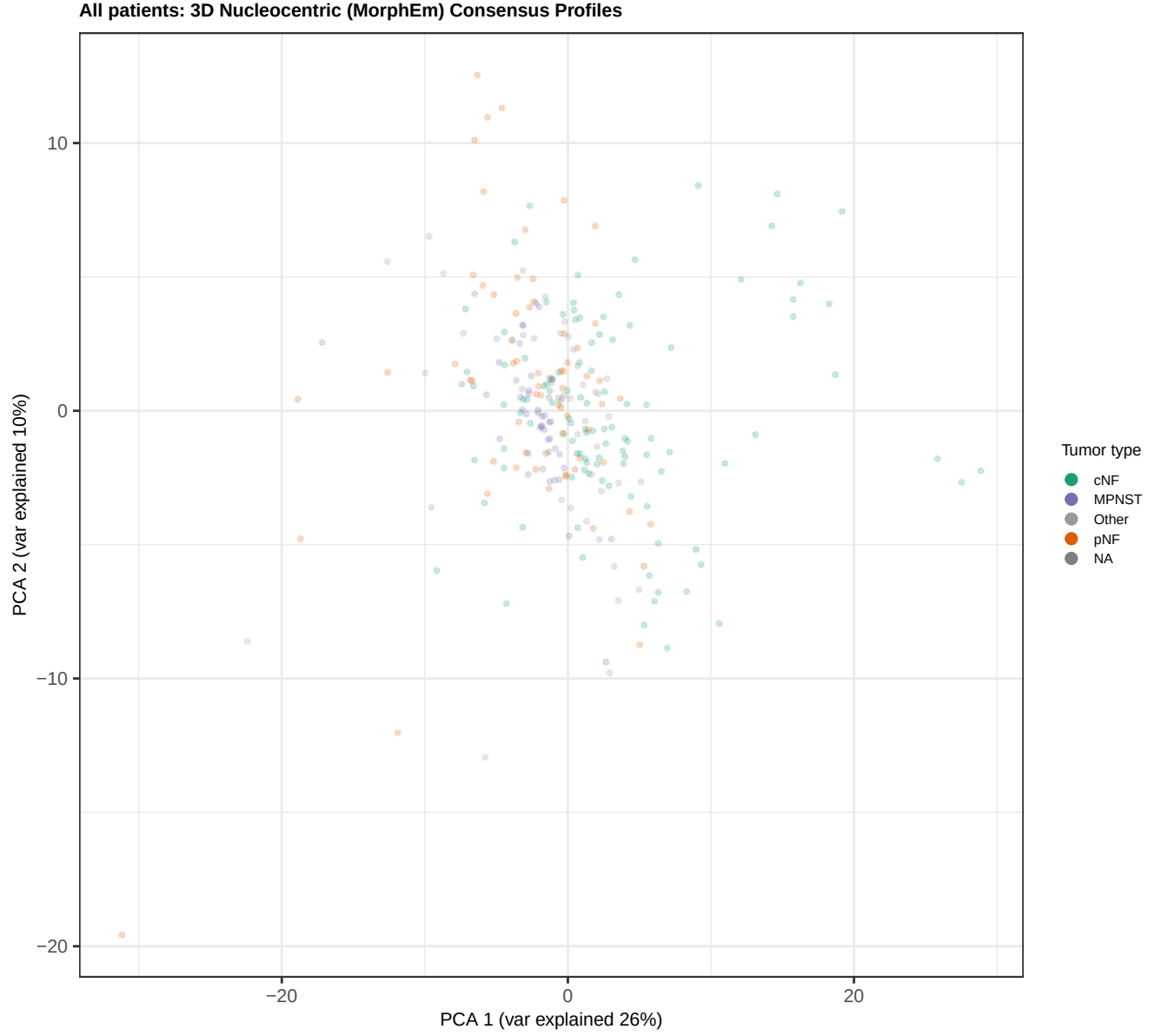

All patients: 3D Nucleocentric (MorphEm) Feature Selected Profiles
NF0014_T1
NF0014_T2
NF0016_T1
NF0018_T6
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1
NA
PCA 2 (var explained 5%)
PCA 1 (var explained 6%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA

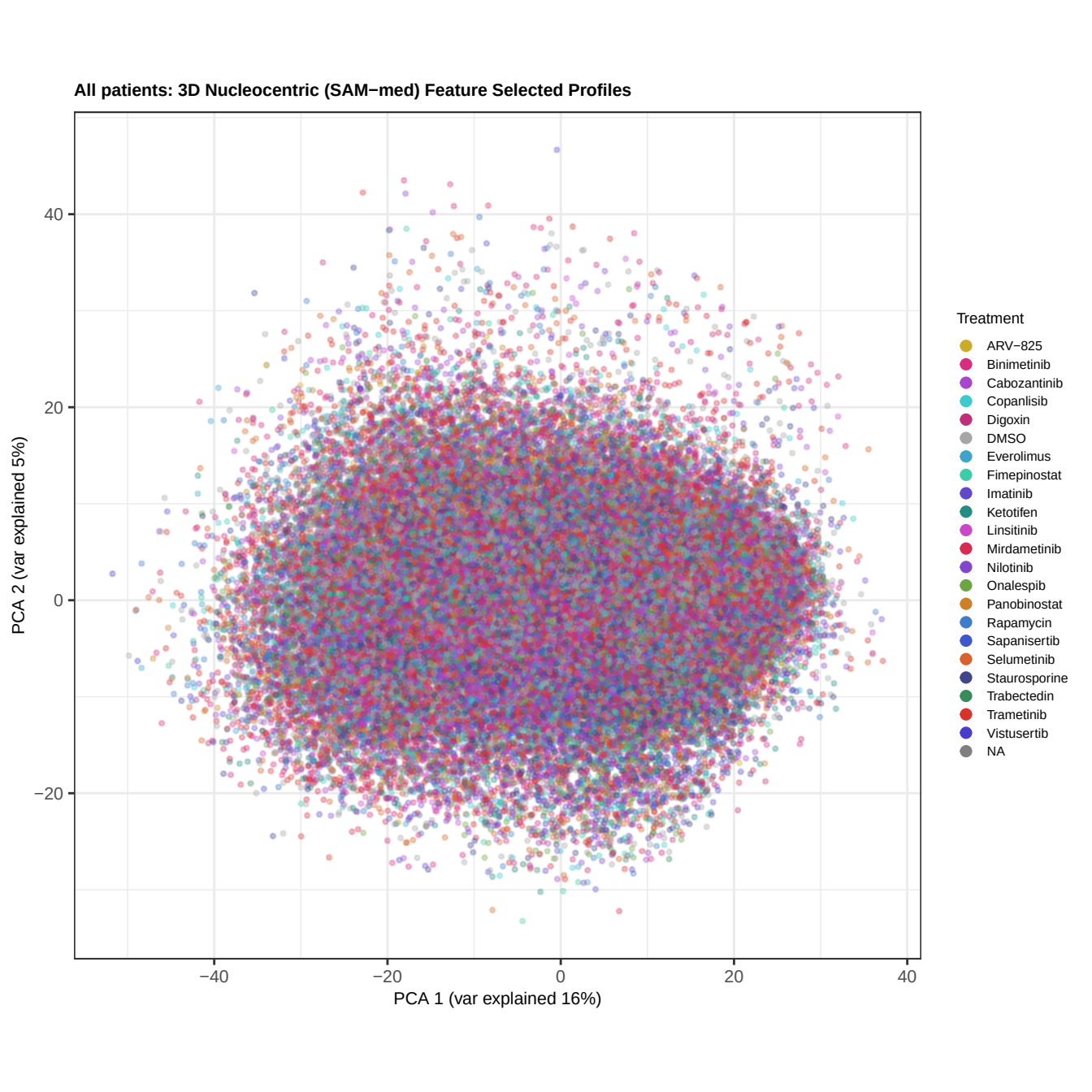

All patients: 3D Nucleocentric (MorphEm) Feature Selected Profiles
cNF
MPNST
Other
pNF
NA
PCA 2 (var explained 5%)
PCA 1 (var explained 6%)
30
20
10
0
−10
−20
−40
−20
0
20
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA

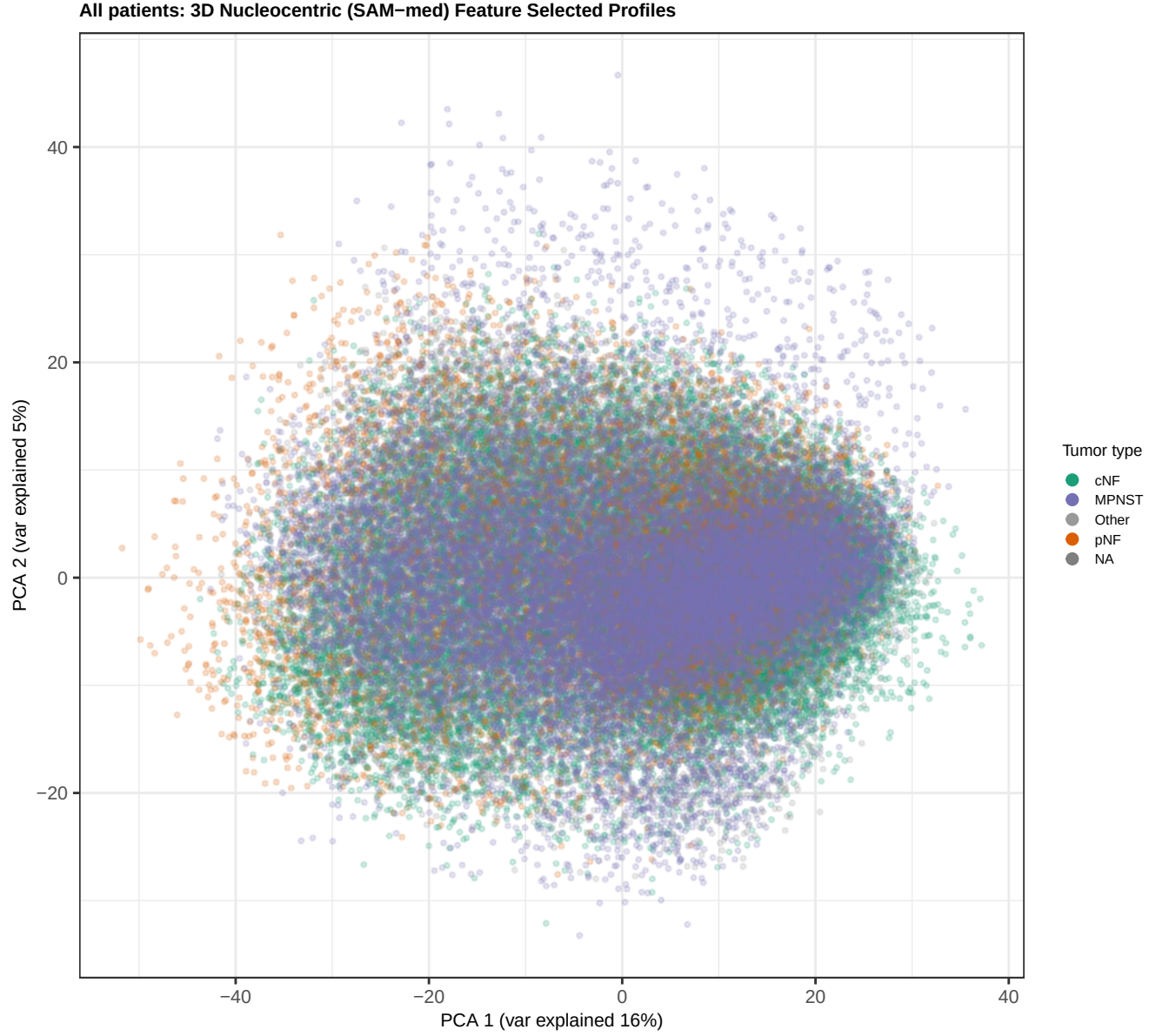

All patients: 3D Nucleocentric (MorphEm) Aggregated Profiles
PCA 2 (var explained 8%)
20
10
0
−10
−20
0
20
PCA 1 (var explained 22%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA

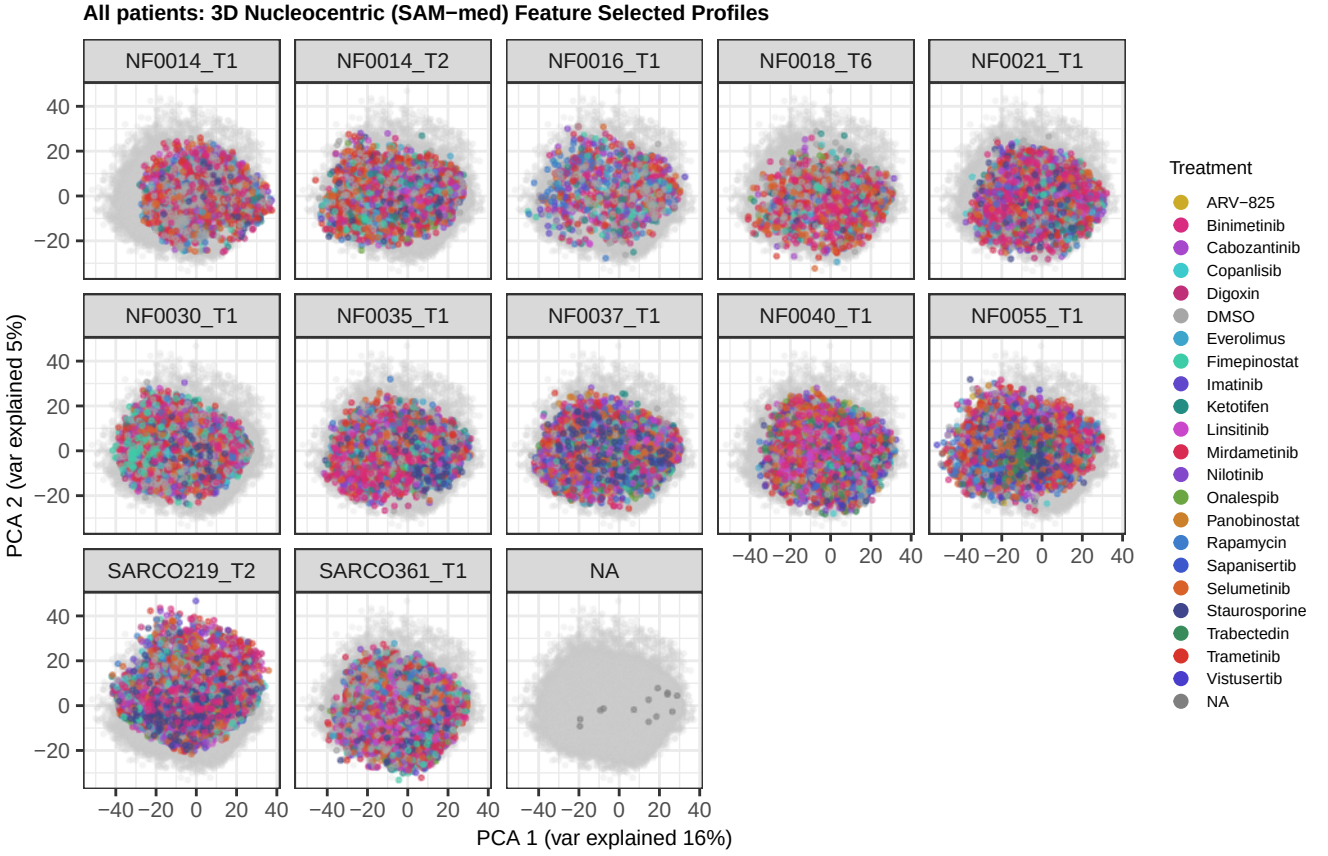

All patients: 3D Nucleocentric (MorphEm) Aggregated Profiles
PCA 2 (var explained 8%)
20
10
0
−10
−20
0
20
PCA 1 (var explained 22%)
Tumor type
cNF
MPNST
Other
pNF
NA

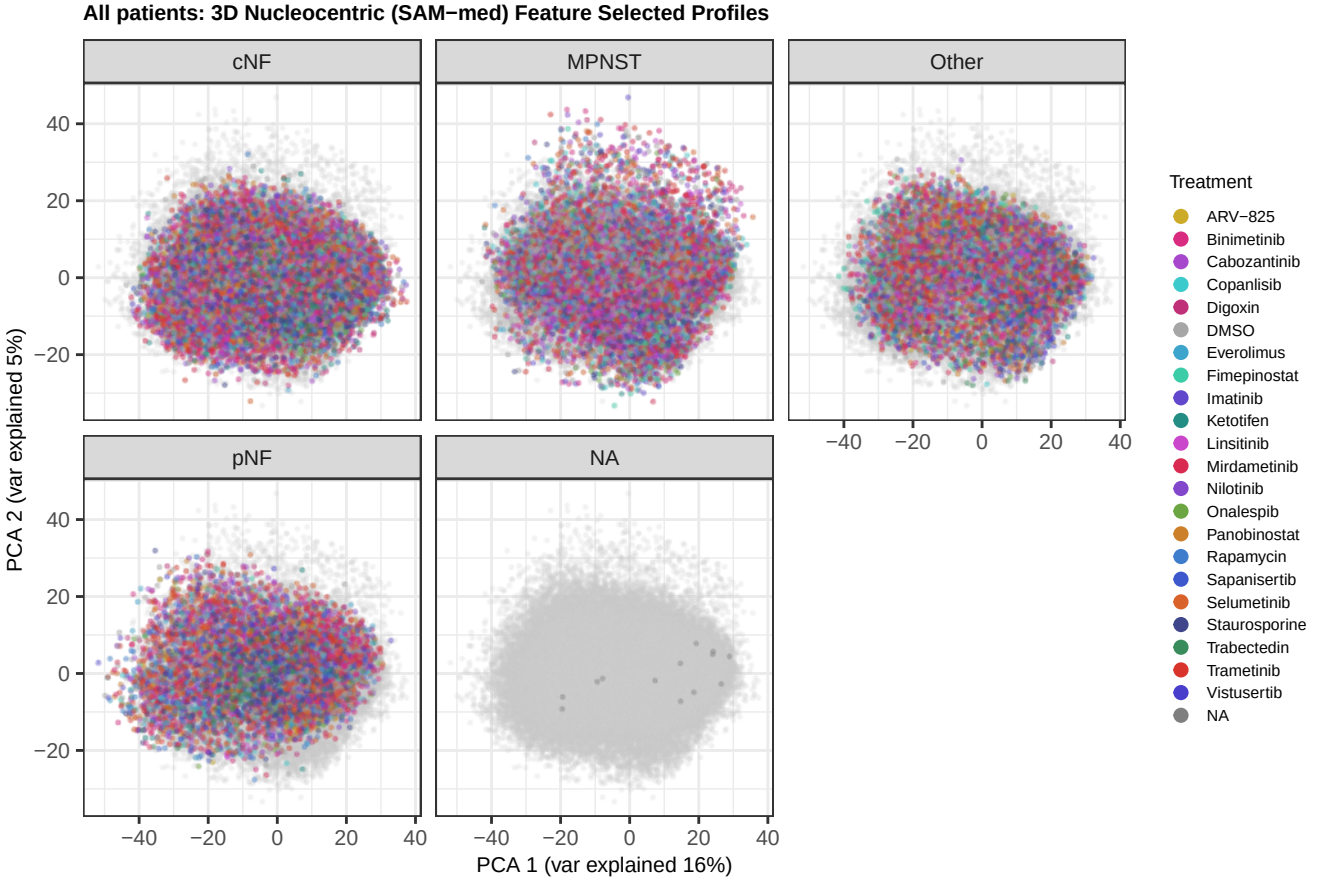

All patients: 3D Nucleocentric (MorphEm) Consensus Profiles
PCA 2 (var explained 10%)
10
0
−10
−20
−20
0
20
PCA 1 (var explained 26%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA

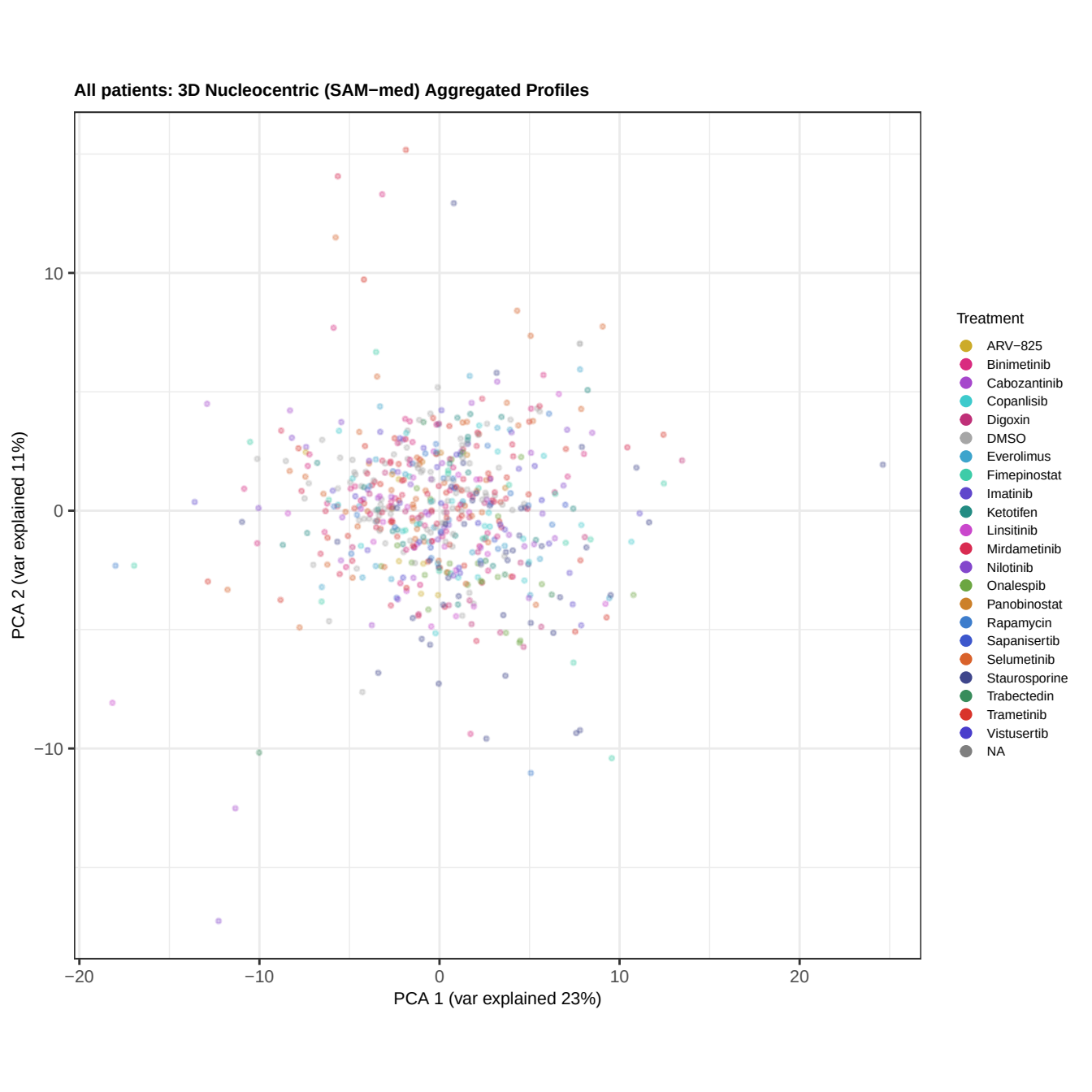

All patients: 3D Nucleocentric (MorphEm) Consensus Profiles
PCA 2 (var explained 10%)
10
0
−10
−20
−20
0
20
PCA 1 (var explained 26%)
Tumor type
cNF
MPNST
Other
pNF
NA

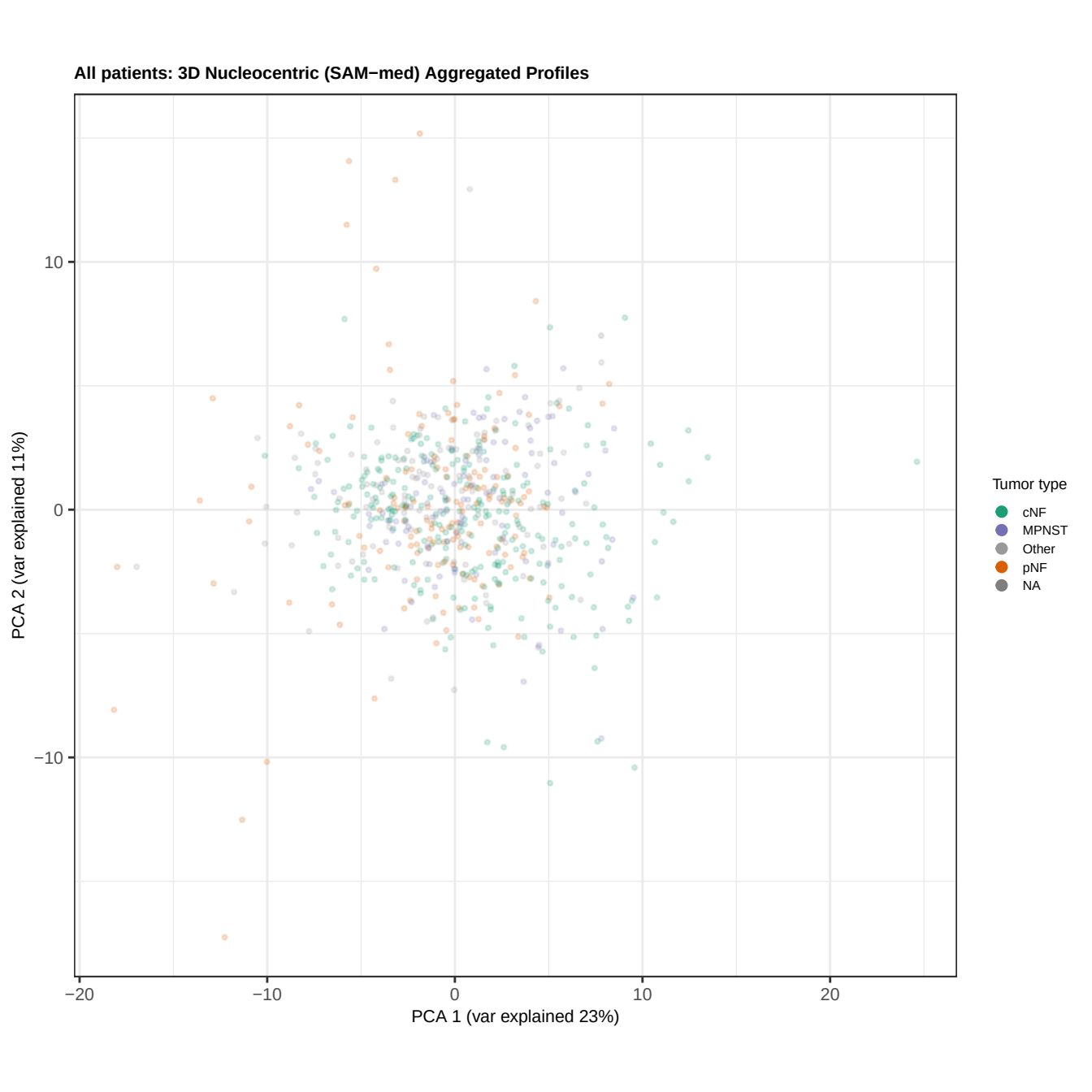

All patients: 3D Nucleocentric (SAM−med) Feature Selected Profiles
PCA 2 (var explained 5%)
PCA 1 (var explained 16%)
40
20
0
−20
−40
−20
0
20
40
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA

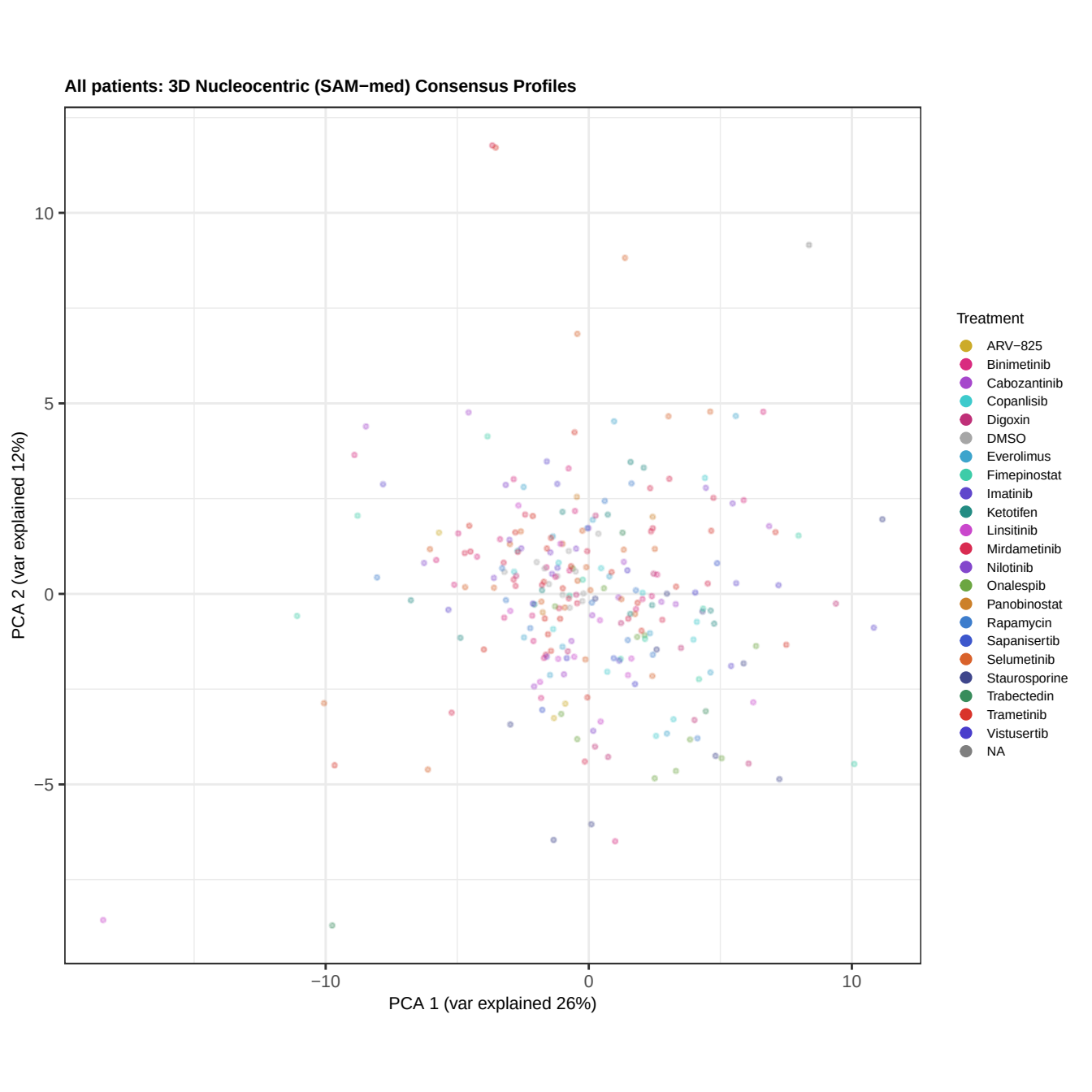

All patients: 3D Nucleocentric (SAM−med) Feature Selected Profiles
PCA 2 (var explained 5%)
PCA 1 (var explained 16%)
40
20
0
−20
−40
−20
0
20
40
Tumor type
cNF
MPNST
Other
pNF
NA

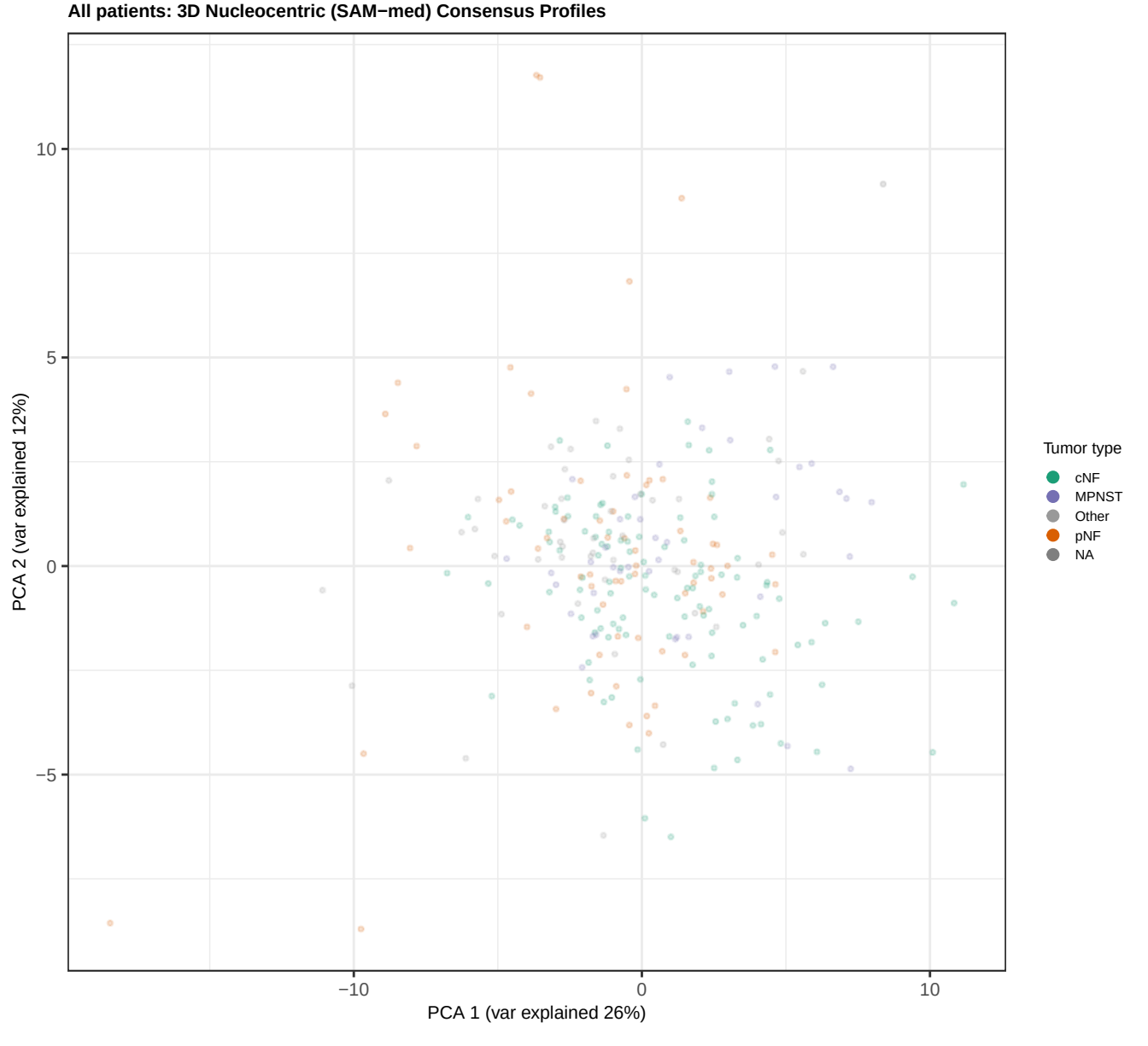

All patients: 3D Nucleocentric (SAM−med) Feature Selected Profiles
NF0014_T1
NF0014_T2
NF0016_T1
NF0018_T6
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1
NA
PCA 2 (var explained 5%)
PCA 1 (var explained 16%)
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA


All patients: 3D Nucleocentric (SAM−med) Feature Selected Profiles
cNF
MPNST
Other
pNF
NA
PCA 2 (var explained 5%)
PCA 1 (var explained 16%)
40
20
0
−20
−40
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA


All patients: 3D Nucleocentric (SAM−med) Aggregated Profiles
PCA 2 (var explained 11%)
PCA 1 (var explained 23%)
10
0
−10
−20
−10
0
10
20
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA


All patients: 3D Nucleocentric (SAM−med) Aggregated Profiles
PCA 2 (var explained 11%)
10
0
−10
−20
−10
0
10
20
PCA 1 (var explained 23%)
Tumor type
cNF
MPNST
Other
pNF
NA


All patients: 3D Nucleocentric (SAM−med) Consensus Profiles
PCA 2 (var explained 12%)
PCA 1 (var explained 26%)
10
5
0
−5
−10
0
10
Treatment
ARV−825
Binimetinib
Cabozantinib
Copanlisib
Digoxin
DMSO
Everolimus
Fimepinostat
Imatinib
Ketotifen
Linsitinib
Mirdametinib
Nilotinib
Onalespib
Panobinostat
Rapamycin
Sapanisertib
Selumetinib
Staurosporine
Trabectedin
Trametinib
Vistusertib
NA


All patients: 3D Nucleocentric (SAM−med) Consensus Profiles
PCA 2 (var explained 12%)
10
5
0
−5
−10
0
10
PCA 1 (var explained 26%)
Tumor type
cNF
MPNST
Other
pNF
NA